

THE UNIQUE OFF-POLICY ADMISSIBILITY OF KL DIVERGENCE IN DIRECT PREFERENCE OPTIMIZATION

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ABSTRACT

Direct preference optimization (DPO) methods almost universally regularize the learned policy against a reference using Kullback–Leibler (KL) divergence. We show that this choice is structural rather than conventional. Among a broad class of separable strictly convex regularizers, KL is the unique divergence whose implicit reward representation can be computed from transition tuples (s, a, s') alone, without requiring additional sampling of a next action $a' \sim \pi(\cdot | s')$. We formalize this property as *off-policy admissibility*. For any non-KL regularizer, the implicit reward inherits an inner expectation over next-state actions, forcing a per-state Monte-Carlo estimator whose variance compounds along trajectories. As a result, preference optimization under non-KL regularization becomes intrinsically tied to an on-policy estimation procedure.

To establish this result, we develop a general convex-regularizer framework encompassing both f -divergence and Bregman formulations. Within this framework, we derive: (i) a structural characterization of the optimal reward through the regularizer gradient, (ii) a policy deviation theorem expressing value gaps as discounted sequential Bregman divergences, and (iii) a uniqueness theorem showing that KL is the only off-policy admissible regularizer satisfying natural divergence axioms. Our analysis explains why KL-based DPO remains compatible with logged or asynchronously generated preference data, while non-KL regularizers inherit an additional source of sampling variance absent in KL. Empirically, we confirm that KL recovers its target policy substantially more stably than non-KL alternatives, especially under heterogeneous off-policy preference data.

1 INTRODUCTION

Reinforcement learning (RL) with a KL-divergence penalty against a reference policy π_{ref} has become the dominant paradigm for stable policy optimization, particularly in language-model post-training and reasoning. From maximum-entropy RL and soft actor-critic methods (Schulman et al., 2017; Haarnoja et al., 2018; Geist et al., 2019) to RL from human feedback (RLHF) (Christiano et al., 2017; Ouyang et al., 2022) and Direct Preference Optimization (DPO) (Rafailov et al., 2023; Kim et al., 2025), KL regularization appears almost universally.

A natural question is whether this choice is structural or merely conventional:

Would any divergence regularizer work equally well?

Several works have generalized KL-regularized RL to broader classes of regularizers, including f -divergences (Wang et al., 2024; Belousov & Peters, 2019), Bregman divergences (Tomar et al., 2022; Banerjee et al., 2005), and general convex regularizers (Geist et al., 2019). These formulations admit analogous soft Bellman operators, Legendre–Fenchel duals, and mirror-descent updates. Yet despite this breadth, no concrete advantage of KL over alternative regularizers has been identified.

The advantage we identify concerns off-policy computability of the implicit reward. Modern RLHF pipelines already rely heavily on off-policy data reuse: rollouts are generated asynchronously by stale

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actors (Noukhovitch et al., 2024), filtered or reranked before training (Dong et al., 2023), and reused across many optimization steps (Rafailov et al., 2023). In each case, optimization proceeds on data that the current policy did not generate.

This paper proves that KL is uniquely compatible with this regime within a broad class of regularizers. We consider separable strictly convex divergences between the learned policy and the reference policy, and show that KL is the *only* regularizer whose implicit reward depends solely on the transition tuple (s, a, s') . Every non-KL regularizer introduces an additional inner expectation over next-state actions $a' \sim \pi(\cdot | s')$, which must be estimated by Monte Carlo during training. This additional sampling step makes the estimator intrinsically on-policy, and its variance compounds along trajectories.

The distinction mirrors the classical Q -learning versus SARSA dichotomy (Watkins & Dayan, 1992; Rummerly & Niranjan, 1994). Like Q -learning, KL avoids sampling the next action. Every other regularizer requires it, as SARSA does. We formalize this property as *off-policy admissibility*. Because KL requires no next-action sample, its implicit reward remains valid on logged trajectories generated by arbitrary behavior policies. Our experiments show that this yields substantially more stable policy recovery under heterogeneous off-policy preference data.

We summarize the main contributions of this paper as follows:

- **Implicit reward representation for general convex regularization.** We extend the DPO implicit-reward construction from f -divergences to arbitrary separable strictly convex regularizers.
- **KL uniqueness via off-policy admissibility.** We identify *off-policy admissibility*, the property that the implicit reward is computable from (s, a, s') without sampling $a' \sim \pi(\cdot | s')$, and prove that KL is the unique regularizer satisfying it.
- **Empirical consequences.** Under fixed compute, the additional Monte-Carlo variance induced by non-KL regularizers produces a measurable trained-policy quality gap, especially in heterogeneous off-policy settings.

2 RELATED WORK

KL-regularized policy optimization. KL has long served as the regularizer of choice across many fields, including trust-region policy optimization (Schulman et al., 2015), imitation learning (Ho & Ermon, 2016), and post-training of language models (Stiennon et al., 2020). In RL specifically, it has been the standard regularizer since the maximum-entropy framework was unified with policy gradients (Schulman et al., 2017; Haarnoja et al., 2018). The regularizer enters the soft Bellman operator and yields a closed-form softmax optimal policy. A body of work has analyzed why KL has such strong convergence and error-propagation guarantees (Vieillard et al., 2020). We contribute to this line a structural fact, that among separable divergence regularizers only KL admits a closed-form implicit reward computable from the transition tuple (s, a, s') without an extra next-state sample.

General convex regularizers and Bregman divergence in RL. To extend KL-regularized RL while retaining a closed-form policy structure, several lines of work introduced broader convex penalties whose Legendre–Fenchel dual still admits an explicit soft-policy form. Geist et al. (2019) develops the general theory of regularized Markov decision processes, generalizing entropy and KL penalties to arbitrary strictly convex regularizers via the Legendre–Fenchel transform. Belousov & Peters (2019) extends the analysis to α -divergences and identifies the actor-critic pair arising from each f -divergence penalty. Tomar et al. (2022) introduces mirror-descent policy optimization, recasting trust-region updates as proximal steps under a Bregman divergence.

Direct preference optimization and its descendants. Rafailov et al. (2023) introduce DPO in the bandit setting, observing that the KL-regularized RLHF objective admits a closed-form implicit reward $\alpha \log(\pi_\theta / \pi_{\text{ref}})$ that can be matched directly to preferences. Wang et al. (2024) generalize this construction to arbitrary f -divergence regularizers, still in the bandit setting. Cho et al. (2025) first establish necessary and sufficient conditions for the existence of a closed-form implicit reward in the MDP setting under the maximum-entropy framework, and Kim et al. (2025) extend DPO to LLM reasoning tasks by presenting analogous conditions for the MDP setting under the KL-regularized framework. Our contribution generalizes both lines of work in two directions: we lift the analysis

from the f -divergence parameterization to any separable strictly convex regularizer Ω (Theorem 2), and we prove that within this broader class the closed-form implicit reward is *uniquely* characterized by KL via off-policy admissibility.

3 BACKGROUND AND NOTATION

Throughout the paper, we work in the general convex regularizer framework. We first characterize the optimal reward through the gradient of the regularizer Ω , and then express the value gap between any policy and the α -optimal one as a discounted sum of Bregman divergences. From the structural form of the reward we extract the implicit reward representation that gives rise to off-policy admissibility, and we close by proving that KL is the unique regularizer for which this property holds. Each result is stated for a general Ω and then specialized to the f -divergence and KL cases.

Notation. We consider a Markov decision process with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $\mathbb{P}(s'|s, a)$, discount $\gamma \in (0, 1]$, and unregularized reward $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$. The regularization is set by a fixed reference policy π_{ref} and a temperature $\alpha > 0$. We parametrize policies as $\pi_\theta(\cdot|s) \in \Delta(\mathcal{A})$ and write π when θ is not the focus. The regularizer $\Omega(\cdot; \pi_{\text{ref}}) : \Delta(\mathcal{A}) \rightarrow \mathbb{R}$ is strictly convex in its first argument. In language-model post-training, states are token prefixes, actions are next tokens, and transitions are deterministic.

Our running specialization is the f -divergence,

$$D_f(\pi \| \pi_{\text{ref}}) := \sum_{a \in \mathcal{A}} \pi_{\text{ref}}(a) f\left(\frac{\pi(a)}{\pi_{\text{ref}}(a)}\right), \quad (1)$$

for strictly convex f with $f(1) = 0$. Our ablation uses the seven regularizers in Table 1, six of them f -divergences. We write the α -divergence parameter as α_{div} to avoid clashing with the temperature α . In this parameterization, the α -divergence interpolates from KL ($\alpha_{\text{div}} \rightarrow 0$) to reverse KL ($\alpha_{\text{div}} \rightarrow 1$).

Uniform reference. When π_{ref} is uniform on $\mathcal{A}$ with $n = |\mathcal{A}|$, the f -divergence reduces to a per-state entropy penalty. KL ($f(t) = t \log t$) gives $-H(\pi) + \log n$, recovering maximum-entropy RL, while the α -divergence gives

$$D_f(\pi \| \text{unif}) = -n^{q-1} S_q(\pi) + \text{const}, \quad q := 1 - \alpha_{\text{div}},$$

with the Tsallis q -entropy $S_q(\pi) = \frac{1}{q-1}(1 - \sum_a \pi(a)^q)$. Its endpoints recover Shannon entropy as $q \rightarrow 1$ and, for Pearson χ^2 , the sparsemax penalty $n\|\pi\|_2^2 - 1$ at $q = 2$ (Lee et al., 2018).

Ω -regularized objective and the α -optimal policy. The Ω -regularized RLHF objective and its associated soft Bellman operator are

$$J(\pi) := \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) - \alpha \Omega(\pi(\cdot|s_t); \pi_{\text{ref}}(\cdot|s_t))) \right], \quad (2)$$

$$\mathcal{T}_\Omega V(s) := \max_{\pi(\cdot|s) \in \Delta(\mathcal{A})} \mathbb{E}_{a \sim \pi} [r(s, a) + \gamma \mathbb{E}_{s' \sim \mathbb{P}(\cdot|s, a)} [V(s')]] - \alpha \Omega(\pi(\cdot|s); \pi_{\text{ref}}(\cdot|s)). \quad (3)$$

The unique fixed point of the soft Bellman operator $\mathcal{T}_\Omega$ in Equation 3 is the regularized optimal value V^* , and the associated action-value is $Q^*(s, a) := r(s, a) + \gamma \mathbb{E}_{s' \sim \mathbb{P}(\cdot|s, a)} [V^*(s')]$. The α -optimal policy w.r.t. Ω , denoted π^* , is the per-state argmax inside Equation 3 at $V = V^*$,

$$\pi^*(\cdot|s) := \arg \max_{\pi(\cdot|s) \in \Delta(\mathcal{A})} \mathbb{E}_{a \sim \pi} [Q^*(s, a)] - \alpha \Omega(\pi(\cdot|s); \pi_{\text{ref}}(\cdot|s)),$$

and is uniquely determined when Ω is strictly convex in its first argument. We also write V^π , Q^π for the regularized value and action-value of an arbitrary policy π , defined by

$$V^\pi(s) := \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t (r(s_t, a_t) - \alpha \Omega(\pi(\cdot|s_t); \pi_{\text{ref}}(\cdot|s_t))) \mid s_0 = s \right],$$

$$Q^\pi(s, a) := r(s, a) + \gamma \mathbb{E}_{s' \sim \mathbb{P}(\cdot|s, a)} [V^\pi(s')].$$

The functional gradient $\nabla_{\pi}\Omega(\cdot; \pi_{\text{ref}}) : \Delta(\mathcal{A}) \rightarrow \mathbb{R}^{|\mathcal{A}|}$ is defined componentwise by $[\nabla_{\pi}\Omega(\pi; \pi_{\text{ref}})]_a := \partial\Omega(\pi; \pi_{\text{ref}})/\partial\pi(a)$, and the convex conjugate is $\Omega^*(y; \pi_{\text{ref}}) := \sup_{\pi \in \Delta(\mathcal{A})} \langle \pi, y \rangle - \Omega(\pi; \pi_{\text{ref}})$.

Direct Preference Optimization. Direct Preference Optimization (Rafailov et al., 2023) reparameterizes the KL-regularized RLHF objective in the bandit setting. Preference pairs (x, y_w, y_l) are drawn from a Bradley–Terry oracle on an unknown reward r^* ,

$$\mathbb{P}(y_w \succ y_l \mid x) = \sigma(r^*(x, y_w) - r^*(x, y_l)), \quad \sigma(z) := (1 + e^{-z})^{-1}. \quad (4)$$

The KL-regularized RLHF optimum admits a closed-form implicit reward $\hat{r}_{\theta}(x, y) = \alpha \log(\pi_{\theta}(y|x)/\pi_{\text{ref}}(y|x)) + \alpha \log Z(x)$, where $Z(x)$ is a partition function that depends only on the prompt x . Because $Z(x)$ does not depend on y , it cancels pairwise inside $\hat{r}_{\theta}(x, y_w) - \hat{r}_{\theta}(x, y_l)$. Substituting the cancelled difference into Equation 4 and taking the negative log-likelihood over a preference dataset $\mathcal{D}$ yields the DPO objective,

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\alpha \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \alpha \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right], \quad (5)$$

which fits π_{θ} to preferences without an explicit reward model.

The partition-function cancellation that powers Equation 5 is specific to the bandit setting. In the MDP setting the analogous partition-like term at (s, a) involves the next-state value $V^*(s')$, which varies across trajectories through the transition kernel $\mathbb{P}(\cdot|s, a)$, so it does not cancel inside a trajectory-level preference difference. Many works nonetheless transplant the bandit DPO reward form to the MDP setting without addressing this gap. Cho et al. (2025) and Kim et al. (2025) fix it by characterizing the equivalence class of implicit rewards that share the same α -optimal policy in the MDP setting, parameterized by the reward-shaping freedom $\beta(s)$ of Lemma 1. Our analysis lifts this MDP-setting construction from KL to any separable strictly convex Ω and asks which Ω preserve the closed form without an additional inner expectation at the next state.

3.1 STRUCTURAL FORM OF THE α -OPTIMAL REWARD

This section asks which rewards make a given policy π^* the α -optimal policy under Ω -regularization. Expressing the reward as a closed form in π^* lets us update a candidate policy directly from preference data without ever fitting a separate reward model, the mechanism underlying DPO. The following lemma gives the necessary and sufficient form that the reward and the regularized action-value can take.

Lemma 1 (Structural form of the α -optimal reward). *Assume $\Omega(\cdot; \pi_{\text{ref}})$ is essentially smooth in its first argument, so that the α -optimal policy π^* lies in the interior of $\Delta(\mathcal{A})$ at every state. Then there exists a state-dependent function $\beta : \mathcal{S} \rightarrow \mathbb{R}$ such that for every (s, a) ,*

$$Q^*(s, a) = \alpha [\nabla_{\pi}\Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a + \beta(s), \quad (6)$$

$$r(s, a) = \alpha [\nabla_{\pi}\Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a + \beta(s) - \gamma \mathbb{E}_{s' \sim \mathbb{P}(\cdot|s, a)} [V^*(s')]. \quad (7)$$

Equation 6 identifies the action dependence of Q^* with the gradient of Ω at π^* , up to a state-dependent function $\beta(s)$. Equation 7 is the one-step Bellman backup of the same expression, so the unregularized reward at (s, a) is recovered as the gradient term plus $\beta(s)$ minus the discounted next-state value. The state-dependent term $\beta(s)$ is the reward-shaping freedom of Ng et al. (1999). A proof is given in Appendix A.

f -divergence specialization. For $\Omega(\pi; \pi_{\text{ref}}) = D_f(\pi \| \pi_{\text{ref}})$, the functional gradient evaluates pointwise to $[\nabla_{\pi}\Omega(\pi; \pi_{\text{ref}})]_a = f'(\pi(a)/\pi_{\text{ref}}(a))$, and Equation 6 becomes

$$Q^*(s, a) = \alpha f' \left(\frac{\pi^*(a|s)}{\pi_{\text{ref}}(a|s)} \right) + \beta(s).$$

The KL case $f(t) = t \log t$ gives $f'(t) = \log t + 1$. The additive constant absorbs into β through reward shaping, leaving

$$Q^*(s, a) = \alpha \log \frac{\pi^*(a|s)}{\pi_{\text{ref}}(a|s)} + \beta(s)$$

and the implicit reward $r(s, a) = \alpha \log(\pi^*(a|s)/\pi_{\text{ref}}(a|s)) + \beta(s) - \gamma \mathbb{E}[\beta(s')]$ as in Kim et al. (2025). Taking π_{ref} uniform on $\mathcal{A}$ reduces $D_{\text{KL}}(\pi \| \pi_{\text{ref}})$ to negative entropy up to a constant, so the framework collapses to the maximum-entropy setting of Haarnoja et al. (2018); Cho et al. (2025).

Table 1: Regularizers $\Omega(\pi; \pi_{\text{ref}})$ used in our ablation. For f -divergence rows, $\Omega(\pi; \pi_{\text{ref}}) = \sum_a \pi_{\text{ref}}(a) f(u)$ with $u := \pi(a)/\pi_{\text{ref}}(a)$. Squared Euclidean is a Bregman regularizer with potential $\Omega(\pi) = \frac{1}{2} \|\pi\|^2$.

Name	$\Omega(\pi; \pi_{\text{ref}})$	Generator $f(u)$
KL ($\alpha_{\text{div}} = 0$)	$\sum_a \pi(a) \log \frac{\pi(a)}{\pi_{\text{ref}}(a)}$	$u \log u$
Reverse KL ($\alpha_{\text{div}} = 1$)	$\sum_a \pi_{\text{ref}}(a) \log \frac{\pi_{\text{ref}}(a)}{\pi(a)}$	$-\log u$
α -div. ($\alpha_{\text{div}} \in (0, 1)$)	$\frac{\sum_a \pi_{\text{ref}}(a) [u^{1-\alpha_{\text{div}}} - (1 - \alpha_{\text{div}})u - \alpha_{\text{div}}]}{\alpha_{\text{div}}(\alpha_{\text{div}} - 1)}$	$\frac{u^{1-\alpha_{\text{div}}} - (1 - \alpha_{\text{div}})u - \alpha_{\text{div}}}{\alpha_{\text{div}}(\alpha_{\text{div}} - 1)}$
Jensen–Shannon	$\frac{1}{2} \sum_a \left[\pi(a) \log \frac{2\pi(a)}{\pi(a) + \pi_{\text{ref}}(a)} + \pi_{\text{ref}}(a) \log \frac{2\pi_{\text{ref}}(a)}{\pi(a) + \pi_{\text{ref}}(a)} \right]$	$\frac{1}{2} \left[u \log u - (u + 1) \log \frac{u+1}{2} \right]$
Pearson χ^2	$\sum_a \pi_{\text{ref}}(a) \left(\frac{\pi(a)}{\pi_{\text{ref}}(a)} - 1 \right)^2$	$(u - 1)^2$
Sq. Hellinger	$\sum_a \pi_{\text{ref}}(a) \left(\sqrt{\frac{\pi(a)}{\pi_{\text{ref}}(a)}} - 1 \right)^2$	$(\sqrt{u} - 1)^2$
Sq. Euclidean	$\frac{1}{2} \sum_a (\pi(a) - \pi_{\text{ref}}(a))^2$	—

3.2 POLICY DEVIATION THEOREM

Lemma 1 pins the reward and the action-value to the α -optimal policy at every state. We now turn from π^* to an arbitrary policy π and ask how its value compares. The next theorem expresses the value gap $Q^* - Q^\pi$ as a discounted sum of Bregman divergences associated with Ω , accumulated along trajectories generated under π .

We first recall the *functional Bregman divergence* of $\Omega(\cdot; \pi_{\text{ref}})$ on the simplex, at any two policies $\pi, \pi^* \in \Delta(\mathcal{A})$,

$$B_\Omega(\pi \| \pi^*) := \Omega(\pi; \pi_{\text{ref}}) - \Omega(\pi^*; \pi_{\text{ref}}) - \langle \nabla_\pi \Omega(\pi^*; \pi_{\text{ref}}), \pi - \pi^* \rangle. \quad (8)$$

Here, the Bregman divergence measures the gap between $\Omega(\pi; \pi_{\text{ref}})$ and the first-order Taylor expansion of Ω about π^* evaluated at π . It is nonnegative whenever Ω is convex and is central to the analysis of regularized MDPs (Geist et al., 2019). Among the seven regularizers in Table 1, squared Euclidean is the canonical Bregman regularizer that does not factor as an f -divergence, and the general convex regularizer setting of this section captures it on the same footing as the six f -divergence cases.

Theorem 1 (General convex regularizer policy deviation). *For any policy π and any (s, a) ,*

$$Q^*(s, a) - Q^\pi(s, a) = \alpha \bar{B}_\Omega(\pi \| \pi^*; s, a), \quad (9)$$

where the sequential Bregman divergence of Ω accumulates the per-state Bregman divergence in Equation 8 along discounted trajectories generated from (s, a) under π ,

$$\bar{B}_\Omega(\pi \| \pi^*; s, a) := \mathbb{E}_{\tau \sim \mathbb{P}_{\pi, a}^s} \left[\sum_{l \geq 0} \gamma^l B_\Omega(\pi(\cdot | s_l) \| \pi^*(\cdot | s_l)) \right]. \quad (10)$$

Although Ω enters Equation 2 only as a state-wise additive penalty, it induces a Bregman metric on the action-value space, and the value gap is measured against that metric. Theorem 1 gives a clean accounting of the suboptimality of π relative to π^* . Each step taken under π away from π^* contributes the per-state Bregman divergence $B_\Omega(\pi(\cdot | s_l) \| \pi^*(\cdot | s_l))$, and these contributions accumulate with discount γ^l along the trajectory. The gap is invariant under the reward-shaping freedom of Lemma 1, since B_Ω depends only on Ω and the two policies and is therefore independent of β . A proof is given in Appendix A.

f -divergence specialization. Two recoveries follow when the regularizer is specialized within the f -divergence parameterization $\Omega(\pi; \pi_{\text{ref}}) = D_f(\pi \parallel \pi_{\text{ref}})$. The KL case $f(t) = t \log t$ reduces the per-state Bregman divergence to $B_\Omega(\pi \parallel \pi^*) = D_{\text{KL}}(\pi \parallel \pi^*)$, and Equation 9 becomes

$$Q^*(s, a) - Q^\pi(s, a) = \alpha \mathbb{E}_{\tau \sim \mathbb{P}_{\pi, a}} \left[\sum_{l \geq 0} \gamma^l D_{\text{KL}}(\pi(\cdot | s_l) \parallel \pi^*(\cdot | s_l)) \right],$$

which is the KL-regularized policy-deviation form of Kim et al. (2025, Theorem 5.4). Further taking π_{ref} to be uniform on $\mathcal{A}$ reduces $D_{\text{KL}}(\pi \parallel \pi_{\text{ref}})$ to negative entropy up to an additive constant, placing the framework in the maximum-entropy setting studied by Cho et al. (2025).

3.3 IMPLICIT REWARD AND OFF-POLICYNESS

Lemma 1 expresses the unregularized reward in terms of the α -optimal policy π^* and value V^* . We restate that identity here as the *implicit reward representation*,

$$r(s, a) = \alpha [\nabla_\pi \Omega(\pi^*(\cdot | s); \pi_{\text{ref}}(\cdot | s))]_a + \beta(s) - \gamma \mathbb{E}_{s' \sim \mathbb{P}(\cdot | s, a)} [V^*(s')]. \quad (11)$$

To evaluate Equation 11 at training time, the next-state value $V^*(s')$ must itself be expressed in terms of π^* . Substituting the structural form for Q^* from Equation 6 into the soft Bellman identity for $V^*(s')$, and pulling $\beta(s')$ outside the expectation over a' , gives a compact form.

$$\begin{aligned} V^*(s') &= \mathbb{E}_{a' \sim \pi^*(\cdot | s')} [Q^*(s', a')] - \alpha \Omega(\pi^*(\cdot | s'); \pi_{\text{ref}}(\cdot | s')) \\ &= \alpha \left(\underbrace{\mathbb{E}_{a' \sim \pi^*(\cdot | s')} [\nabla_\pi \Omega(\pi^*(\cdot | s'); \pi_{\text{ref}}(\cdot | s'))]_{a'}}_{a' \text{-dependent term}} - \Omega(\pi^*(\cdot | s'); \pi_{\text{ref}}(\cdot | s')) \right) + \beta(s') \\ &=: \alpha C_\Omega(\pi^*(\cdot | s'); \pi_{\text{ref}}(\cdot | s')) + \beta(s'), \end{aligned} \quad (12)$$

where the policy-dependent functional C_Ω is defined as

$$C_\Omega(\pi; \pi_{\text{ref}}) := \mathbb{E}_{a \sim \pi} [\nabla_\pi \Omega(\pi; \pi_{\text{ref}})]_a - \Omega(\pi; \pi_{\text{ref}}). \quad (13)$$

The middle line of the derivation makes explicit that $V^*(s')$ inherits an inner expectation over $a' \sim \pi^*(\cdot | s')$ at every reachable next state, which the compact C_Ω shorthand on the third line absorbs but does not eliminate. Substituting the compact form back into Equation 11 yields the four-term reward expansion

$$r(s, a) = \alpha [\nabla_\pi \Omega(\pi^*(\cdot | s); \pi_{\text{ref}}(\cdot | s))]_a + \beta(s) - \gamma \alpha \mathbb{E}_{s'} [C_\Omega(\pi^*(\cdot | s'); \pi_{\text{ref}}(\cdot | s'))] - \gamma \mathbb{E}_{s'} [\beta(s')]. \quad (14)$$

For a general Ω the functional C_Ω has no closed form, so at training time it is replaced by the Monte-Carlo estimator

$$\hat{C}_\Omega^{(n)}(\pi; \pi_{\text{ref}}) := \frac{1}{n} \sum_{i=1}^n [\nabla_\pi \Omega(\pi; \pi_{\text{ref}})]_{a_i} - \Omega(\pi; \pi_{\text{ref}}), \quad a_i \sim \pi, \quad (15)$$

which draws n fresh actions at every reachable next state, with $a_i = a'_i \sim \pi^*(\cdot | s')$ when applied at s' . The empirical implicit reward $\hat{r}_\theta$ that trajectory-DPO fits to preferences is read off Equation 14 with $\hat{C}_\Omega^{(n)}$ in place of C_Ω , so it inherits these next-state samples through the third term. Whether they can be avoided is the central concern of this subsection.

Off-policy admissibility. The third term in Equation 14, $\gamma \alpha \mathbb{E}_{s'} [C_\Omega(\pi^*(\cdot | s'); \pi_{\text{ref}}(\cdot | s'))]$, is the obstruction to a tractable implicit-reward representation. Whether it forces next-state sampling depends entirely on how C_Ω varies with the next-state action a' . When $C_\Omega(\pi(\cdot | s'); \pi_{\text{ref}}(\cdot | s'))$ depends on a' through $\pi(\cdot | s')$, the Monte-Carlo estimator $\hat{C}_\Omega^{(n)}$ of Equation 15 is unavoidable, because evaluating the term draws $a' \sim \pi^*(\cdot | s')$ at every reachable next state s' , on top of the transition sample $s' \sim \mathbb{P}(\cdot | s, a)$. The implicit reward then inherits an extra layer of Monte-Carlo

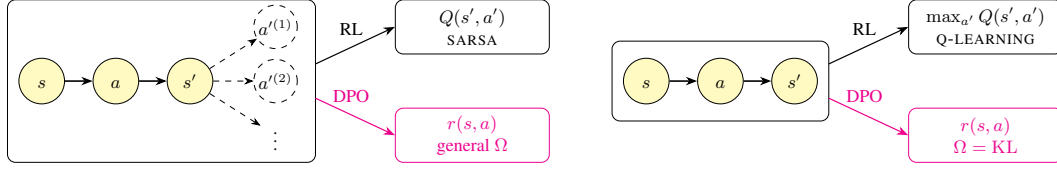


Figure 1: Off-policy admissibility schematic, cast as the Q -learning vs. SARSA distinction. **(Left)** For a general Ω , the implicit reward $r(s, a)$ behaves like SARSA’s bootstrap target $Q(s', a')$: both are evaluated at a transition (s, a, s') only after drawing an inner Monte-Carlo sample $a' \sim \pi^*(\cdot|s')$ at the next state (dashed). **(Right)** When $\Omega = \text{KL}$, the implicit reward behaves like Q -learning’s target $\max_{a'} Q(s', a')$: the inner sample vanishes and both quantities are computable from the transition tuple alone.

estimation whose sampling distribution itself depends on π^* . When instead C_Ω is independent of a' , no next-state sample is needed. It reduces to a state-only function $c(s') := C_\Omega(\pi(\cdot|s'); \pi_{\text{ref}}(\cdot|s'))$ at each s' , and the third term groups with the existing next-state β expectation, since both depend on s' only, yielding the three-term form

$$r(s, a) = \alpha [\nabla_{\pi} \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a + \beta(s) - \gamma \mathbb{E}_{s'}[\beta(s') + \alpha c(s')], \quad (16)$$

The implicit reward at (s, a) is then computable from the transition tuple (s, a, s') alone, without any further sample at the next state. We call this property the *off-policy admissibility* of the regularizer Ω , illustrated schematically in Figure 1. The terminology follows the Q -learning versus SARSA distinction. When C_Ω is constant in π , the implicit reward mirrors how Q -learning’s bootstrap target $\max_{a'} Q(s', a')$ requires no sample of a' from any policy, matching panel (b) of Figure 1. When C_Ω varies with π , the implicit reward inherits the inner expectation $\mathbb{E}_{a' \sim \pi^*}(\cdot|s')$ from Equation 12, mirroring SARSA’s bootstrap target $Q(s', a')$ at an actually taken next action $a' \sim \pi(\cdot|s')$, matching panel (a).

3.4 KL UNIQUENESS VIA OFF-POLICY ADMISSIBILITY

Section 3.3 identified off-policy admissibility, the condition that C_Ω is independent of π , as the structural property that makes the implicit reward computable from (s, a, s') tuples alone. We now ask which regularizers Ω satisfy it. The following theorem characterizes KL as the unique element of a class defined by natural divergence axioms.

Theorem 2 (KL uniqueness via off-policy admissibility). *Fix any next state $s' \in \mathcal{S}$ and write $\pi := \pi(\cdot|s')$ and $\pi_{\text{ref}} := \pi_{\text{ref}}(\cdot|s')$. Suppose the per-state regularizer $\Omega(\pi; \pi_{\text{ref}})$ satisfies, for every $\pi_{\text{ref}}(\cdot|s') \in \Delta(\mathcal{A})$:*

- (A1) Coincidence: $\Omega(\pi_{\text{ref}}; \pi_{\text{ref}}) = 0$.
- (A2) Strict positivity: $\Omega(\pi; \pi_{\text{ref}}) > 0$ for every $\pi \neq \pi_{\text{ref}}$.
- (A3) Strict convexity: $\Omega(\cdot; \pi_{\text{ref}})$ is strictly convex on $\Delta(\mathcal{A})$.
- (A4) Off-policy admissibility: the functional $C_\Omega(\pi; \pi_{\text{ref}})$ from Equation 13 is independent of π , equivalently of the next-state action a' .
- (A5) Separability¹: $\Omega(\pi; \pi_{\text{ref}}) = \sum_{a' \in \mathcal{A}} g_{a'}(\pi(a'))$ for some functions $g_{a'} : (0, 1) \rightarrow \mathbb{R}$ that may depend on π_{ref} .

Then at the fixed s' ,

$$\Omega(\pi(\cdot|s'); \pi_{\text{ref}}(\cdot|s')) = c(s') \cdot D_{\text{KL}}(\pi(\cdot|s') \parallel \pi_{\text{ref}}(\cdot|s')), \quad (17)$$

where $c(s') > 0$ depends on s' only through $\pi_{\text{ref}}(\cdot|s')$.

¹We use *separability* in the standard convex-analysis sense (Boyd & Vandenberghe, 2004, §5); Appendix A.5 relates it to decomposability and Csiszár f -divergences.

Equation 17 states that the only Ω satisfying all five axioms is a scalar multiple of KL at each s' , with the scalar $c(s')$ varying across next states only through the reference $\pi_{\text{ref}}(\cdot|s')$. Axioms (A1)–(A3) make Ω a genuine divergence on the simplex. It vanishes at coincidence with the reference, is strictly positive elsewhere, and is strictly convex in its first argument. Axiom (A4) is the off-policy admissibility property identified in Section 3.3 as the requirement for the implicit reward to be computable from (s, a, s') tuples without an inner a' sample. Axiom (A5) restricts to regularizers whose action coordinates contribute additively. This restriction is mild in practice. Standard RLHF regularizers including KL, Rényi divergences, Tsallis entropy, and α -entropy are all separable, and no non-separable convex regularizer is in current RLHF use. A proof is given in Appendix A.

Recent reference-free DPO variants sit naturally inside this axiom class. SimPO (Meng et al., 2024) replaces the explicit reference policy with a length-normalized log-likelihood reward

$$r_{\text{SimPO}}(x, y) = \frac{\beta}{|y|} \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t | y_{<t}, x),$$

which, up to a partition constant, is the KL implicit reward (Equation 19) against a uniform π_{ref} with a per-trajectory rescaling of α . ConfPO (Yoon et al., 2025) extends SimPO by gating the per-token sum with a confidence-based mask $s(y_t) \in \{0, 1\}$ that selects only critical (low-confidence) tokens. This per-token reweighting is absorbed into the shaping freedom $\beta(s_t)$ of Lemma 1. Both variants therefore retain a KL-shaped implicit reward and inherit off-policy admissibility from Theorem 2. By contrast, a hypothetical variant of either construction that swapped KL for a non-KL convex regularizer, for example reverse KL or Jensen–Shannon, would lose off-policy admissibility and require an additional Monte-Carlo sample of a' at every reachable next state during training.

Recovery of the f -divergence case. Take the f -divergence parameterization $\Omega(\pi; \pi_{\text{ref}}) = D_f(\pi \| \pi_{\text{ref}})$ at a fixed next state s' , with $\pi := \pi(\cdot|s')$, $\pi_{\text{ref}} := \pi_{\text{ref}}(\cdot|s')$, and the action index a' ranging over $\mathcal{A}$. The inner-term functional from Equation 13 evaluates to

$$C_{\Omega}(\pi(\cdot|s'); \pi_{\text{ref}}(\cdot|s')) = \mathbb{E}_{a' \sim \pi(\cdot|s')} \left[f' \left(\frac{\pi(a'|s')}{\pi_{\text{ref}}(a'|s')} \right) - \frac{f(\pi(a'|s')/\pi_{\text{ref}}(a'|s'))}{\pi(a'|s')/\pi_{\text{ref}}(a'|s')} \right]. \quad (18)$$

Axiom (A4) thus requires the bracketed integrand, as a function of $t := \pi(a'|s')/\pi_{\text{ref}}(a'|s')$, to be constant in t . Writing this as $t f'(t) - f(t) = ct$ and differentiating gives $t f''(t) = c$, which integrates twice to $f(t) = c(t \log t - t) + dt + e$ for some constants d, e . The condition $f(1) = 0$ fixes $e = c - d$, so $f(t) = ct \log t + (d - c)(t - 1)$. The second summand $(d - c)(t - 1)$ is invisible to D_f : substituting any shift $\kappa(t - 1)$ into the f -divergence formula gives

$$\sum_{a'} \pi_{\text{ref}}(a'|s') \kappa \left(\frac{\pi(a'|s')}{\pi_{\text{ref}}(a'|s')} - 1 \right) = \kappa \left(\sum_{a'} \pi(a'|s') - \sum_{a'} \pi_{\text{ref}}(a'|s') \right) = \kappa(1 - 1) = 0,$$

since both $\pi(\cdot|s')$ and $\pi_{\text{ref}}(\cdot|s')$ are probability distributions on $\mathcal{A}$. Hence only $ct \log t$ contributes, and the only off-policy admissible element of the f -divergence class is therefore

$$D_f(\pi(\cdot|s') \| \pi_{\text{ref}}(\cdot|s')) = c(s') \cdot D_{\text{KL}}(\pi(\cdot|s') \| \pi_{\text{ref}}(\cdot|s')),$$

matching the conclusion of Theorem 2 inside the f -divergence subclass. Setting $f(t) = t \log t$ makes the integrand identically 1, and the implicit reward at the transition (s, a, s') reduces to the closed form

$$r^{\text{KL}}(s, a) = \alpha \log \frac{\pi^*(a|s)}{\pi_{\text{ref}}(a|s)} + \beta(s) - \gamma \mathbb{E}_{s' \sim \mathbb{P}(\cdot|s, a)} [\beta(s')], \quad (19)$$

which depends on no expectation over the next-state action a' . This characterization and the closed-form KL implicit reward in Equation 19 are the f -divergence specialization of Cho et al. (2025) and the KL specialization of Kim et al. (2025), respectively. Theorem 2 extends both to the broader separable strictly convex class governed by axioms (A1)–(A5). The explicit ODE manipulation is given in Appendix A.4.

Remark 1 (Non-separable case). Axiom (A5) restricts attention to separable functionals. Whether the conclusion of Theorem 2 extends to non-separable Ω , that is, whether some non-separable strictly convex divergence might also satisfy axioms (A1) through (A4), remains open. Every convex regularizer in current RLHF practice (KL, Tsallis, α -entropy, Rényi divergences) is separable, so the restriction is not binding in known applications.

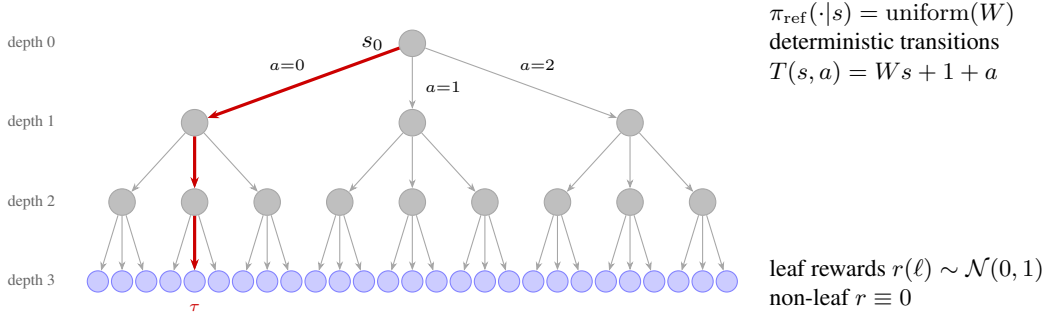


Figure 2: **Toy tree MDP topology** ($W = 3, H = 3, |\mathcal{S}| = 40$). Internal nodes (gray) are non-leaf states. Leaves (blue) carry i.i.d. rewards $r(\ell) \sim \mathcal{N}(0, 1)$. From every non-leaf state, three actions $a \in \{0, 1, 2\}$ deterministically reach the indexed children. The reference policy π_{ref} is uniform. The bold red path shows one trajectory $\tau = (s_0, a_0 = 0, s_1, a_1 = 1, s_2, a_2 = 1, s_3 = \ell_4)$ sampled in trajectory-DPO training. Preferences are generated by a Bradley–Terry oracle on pairs of such trajectories.

4 EXPERIMENTS

This section evaluates three predictions of Theorem 2, framed as the following questions:

1. How does the variance of the inner-term MC estimator $\hat{C}_{\Omega}^{(n)}$ scale with the choice of regularizer, both at a single state and accumulated along a trajectory?
2. Given a fixed per-step MC budget, does trajectory-DPO with KL recover its α -optimal policy and the corresponding implicit reward more accurately than any other Ω ?
3. Does this recovery degrade when the preference data is collected off-policy rather than on-policy from π_{Ω}^* ?

We treat each question in its own subsection after the shared setup.

4.1 SETUP

Environment. Figure 2 shows the toy environment. It is a complete tree MDP of width $W = 3$ and horizon $H = 3$ with $|\mathcal{S}| = 40$ states, deterministic transitions $T(s, a) = Ws + 1 + a$, non-leaf reward 0, and leaf rewards $r(\ell) \sim \mathcal{N}(0, 1)$ drawn once per seed. The reference policy π_{ref} is uniform on every non-leaf state. We instantiate seven regularizers from Table 1: KL, α -divergence at $\alpha_{\text{div}} = 0.1$, Reverse-KL, Jensen–Shannon, squared Hellinger, Pearson χ^2 , and squared Euclidean. Preferences are generated by a Bradley–Terry oracle on raw trajectory returns at temperature $\beta_{\text{or}} = 1$. Training uses tabular logits, the trajectory-DPO loss at $\beta_{\text{DPO}} = 1$, and 5 seeds per cell.

All seven regularizers, including KL, use the same per-step Monte-Carlo budget: at every non-terminal next state, n_{mc} samples are drawn to estimate the inner term C_{Ω} . This keeps the per-step compute matched across Ω and prevents any sampling-budget advantage from confounding the variance comparison. For KL the n_{mc} draws are operationally a no-op ($C_{\text{KL}} \equiv 1$ collapses every sample to the same constant, so n_{mc} has no leverage on the KL gradient), but the samples are still drawn.

Metric. We compare trained policies in *policy space* via the state-occupancy weighted total-variation gap

$$\Delta_{\pi}(\Omega) := \mathbb{E}_{s \sim d^{\pi_{\Omega}^*}} \left[\frac{1}{2} \sum_a |\pi_{\theta}(a | s) - \pi_{\Omega}^*(a | s)| \right]. \quad (20)$$

Two properties make Δ_{π} a fair cross- Ω comparator. First, $\Delta_{\pi} \in [0, 1]$ for every Ω , so the same numerical value carries the same operational meaning across regularizers — unlike a reward-space gap, whose scale depends on Ω ’s gradient and on the reward-shaping freedom of Lemma 1. Second,

$\Delta_\pi(\Omega) = 0$ exactly when $\pi_\theta = \pi_\Omega^*$ on the occupancy support of π_Ω^* , so a small value genuinely reflects policy recovery. The off-policy regime evaluates the same Δ_π on policies trained from uniform-reference rollouts rather than rollouts of π_Ω^* .

A reward-space companion, the equivalent-reward gap $\Delta_r(\Omega) := \mathbb{E}_{(s,a) \sim d^{\pi_\Omega^*}} [|\hat{r}_\theta - r_{\Omega, \text{eq}}^*|]$, is defined together with its underlying canonical reward $r_{\Omega, \text{eq}}^*$ in Appendix B.6, where we report the side-by-side Δ_π/Δ_r comparison. The two stories agree qualitatively, but Δ_r 's scale depends on Ω 's gradient, which makes cross- Ω comparisons noisier; we use Δ_π in the main text.

Per- Ω regularization-weight calibration. Each Ω has a different geometric scale, so at a shared $\alpha = 1$ the peak leaf-reach probability of π_Ω^* varies by an order of magnitude across our roster — for instance Pearson χ^2 at 0.09 and Jensen–Shannon at 0.60 versus KL at 0.19 (mean across 5 leaf-reward draws). To remove this scale artefact from cross- Ω comparisons we sweep α per Ω and pick the calibrated weight α_Ω^* at which $\max_\ell P_{\pi_\Omega^*}(\ell)$ matches KL's anchor at $\alpha = 1$. All Section 4 figures use these calibrated α_Ω^* ; the sweep procedure, the per- Ω values with per-seed range, and the uncalibrated baseline are in Appendix B.5.

4.2 HOW DOES THE VARIANCE OF $\hat{C}_\Omega$ DEPEND ON THE REGULARIZER?

Theorem 2 predicts that the variance of $\hat{C}_\Omega^{(n)}$ is identically zero for $\Omega = \text{KL}$ and strictly positive for every other regularizer in the axiom class. We probe this prediction at two scales, a single state and a full trajectory, both without any training.

Single state. Fix one state with $|\mathcal{A}| = 100$ actions, set π_{ref} to uniform, and draw a policy π as a softmax of standard-Gaussian logits scaled by 3 (peak mass ≈ 0.24). For each $n \in \{2^2, 2^3, \dots, 2^{10}\}$ we form the n -sample Monte-Carlo estimator of $\mathcal{C}_\Omega(\pi; \pi_{\text{ref}})$ and report its standard deviation across 500 independent draws.

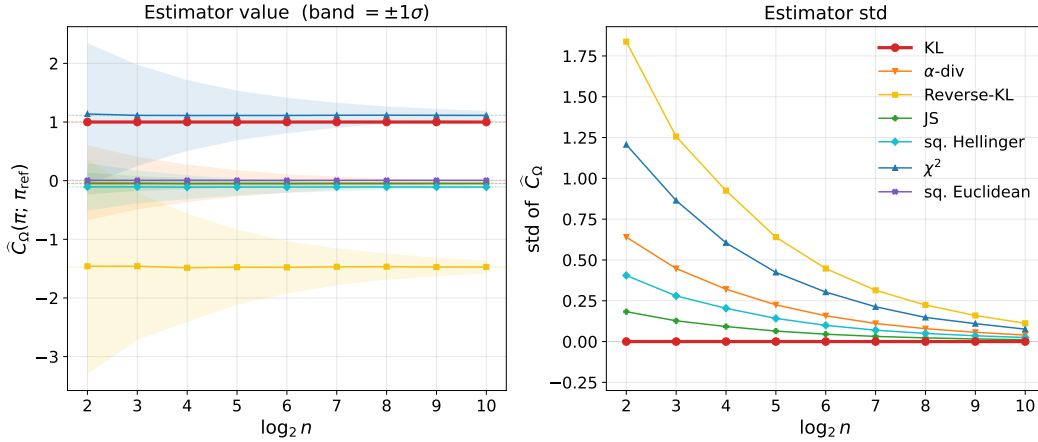


Figure 3: **Single-state inner-term variance vs Monte-Carlo budget n .** Left: estimator value with $\pm 1\sigma$ band; dashed = closed-form exact. Right: estimator std vs n .

Figure 3 confirms the prediction at the single-state scale: KL's estimator collapses to the constant $\mathcal{C}_{\text{KL}} \equiv 1$ at every n , while Reverse-KL, α -divergence, and squared Hellinger each decay as the central-limit $1/\sqrt{n}$. The KL-versus-others gap visible in the right panel is the most direct empirical signature of off-policy admissibility we report.

Along a trajectory. The single-state gap compounds along a rollout. A horizon- H trajectory contributes $H - 1$ independent inner-term draws to the per-trajectory implicit reward, so the per-trajectory standard deviation should grow as $\sqrt{H - 1}$ for non-KL and stay at zero for KL. We measure this directly without training: fix a peaked policy (peakedness 1.0), sample $N_{\text{traj}} = 128$ trajectories per horizon, fix $n_{\text{mc}} = 8$ samples per step, and compute the per-trajectory standard deviation of the inner-term sum over 800 Monte-Carlo realizations at each horizon $H \in \{1, 2, \dots, 8\}$.

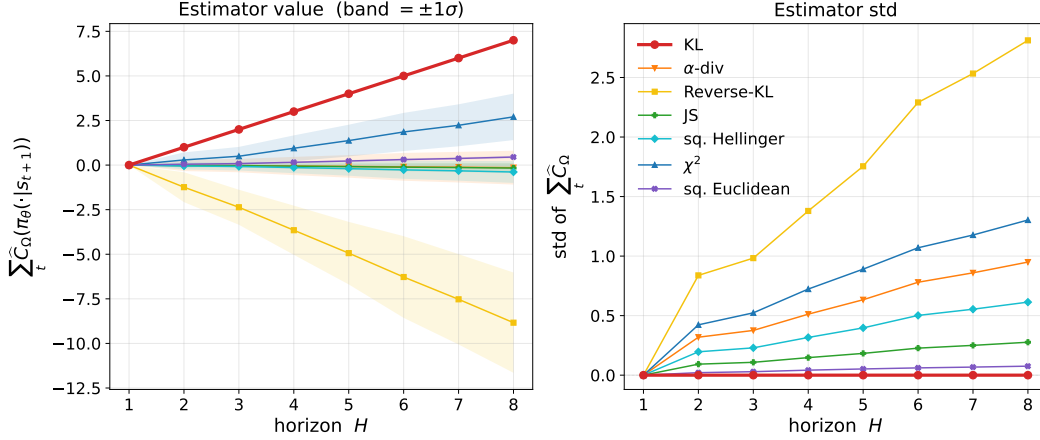


Figure 4: **Trajectory Monte-Carlo compounding vs horizon H .** Left: trajectory-averaged inner-term sum $\sum_t \hat{C}_\Omega(\pi_\theta(\cdot|s_{t+1}))$ with $\pm 1\sigma$ band at $n_{mc} = 8$. Right: per-trajectory std vs H .

Figure 4 shows that KL is identically zero at every horizon, while the six non-KL regularizers grow with H at roughly the predicted $\sqrt{H-1}$ rate. Together with the single-state result, this answers Q1: the variance of $\hat{C}_\Omega$ is identically zero for KL and strictly positive for every other regularizer, and the gap scales as $\sqrt{H-1}$ along the trajectory.

4.3 DOES KL RECOVER ITS α -OPTIMAL POLICY UNDER A FIXED MONTE-CARLO BUDGET?

The variance corollary makes a falsifiable prediction at training time: at any fixed per-step Monte-Carlo budget n_{mc} , trajectory-DPO with $\Omega = \text{KL}$ should recover π_{KL}^* more accurately than any other Ω , with the gap closing only as $n_{mc} \rightarrow \infty$ for those Ω . We test this at $N_{\text{pairs}} = 8000$, sweeping $n_{mc} \in \{1, 2, 4, 8, 16, 32, 64, 128, 256\}$ with 5 seeds per cell and $N_{\text{steps}} = 800$ Adam updates per run. After training we evaluate $\Delta_\pi(\Omega)$ from Equation 20 on the trained π_θ , each Ω at its calibrated α_Ω^* .

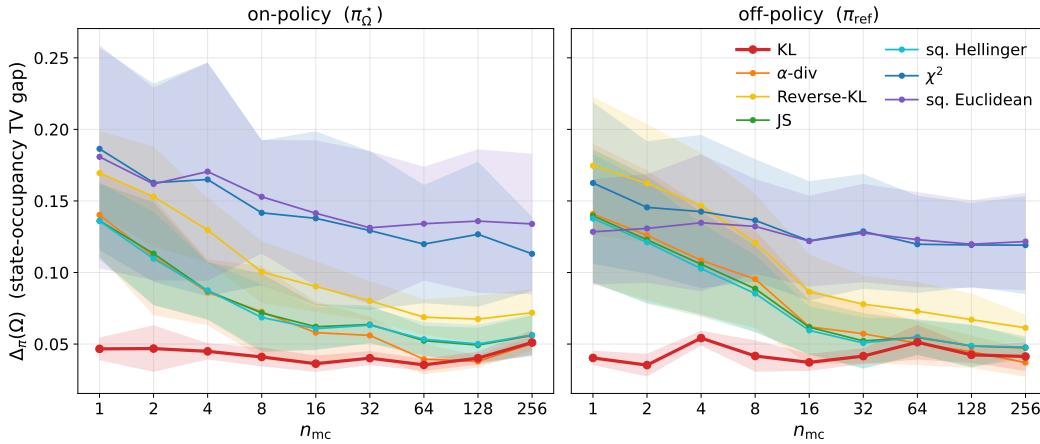


Figure 5: **Trained-policy gap Δ_π versus per-step Monte-Carlo budget n_{mc} .** On-policy (left) and off-policy (right) at $N_{\text{pairs}} = 8000$, calibrated α_Ω^* . Mean and $\pm 1\sigma$ band over 5 seeds; KL drawn thicker.

The on-policy panel of Figure 5 answers Q2 directly. KL stays at $\Delta_\pi \approx 0.04$ across the full n_{mc} range with only seed-level noise, the training-time consequence of the per-state variance corollary verified in Q1. The six non-KL regularizers all start at $\Delta_\pi \in [0.06, 0.19]$ at $n_{mc} = 1$ and converge

monotonically toward KL’s floor by $n_{\text{mc}} = 256$. At $n_{\text{mc}} = 1$ the worst-case KL advantage is roughly $4\times$ (Pearson χ^2 at 0.19 versus KL at 0.05).

Per-leaf anatomy. Figure 6 shows what Δ_π summarizes by integration. For each Ω we plot two per-leaf quantities at horizon $H = 3$ (27 leaves indexed in BFS order on the x -axis): the reach probability $P_\pi(\ell) := \prod_t \pi(a_t|s_t)$ (top sub-panel) and the implicit reward $\hat{R}_\pi(\ell) := \alpha_\Omega^* \sum_t \log(\pi(a_t|s_t)/\pi_{\text{ref}}(a_t|s_t))$ (bottom sub-panel). Under uniform π_{ref} the two are related by $\hat{R}_\pi(\ell) = \alpha_\Omega^* \log P_\pi(\ell) + \text{const}$, so they share the same per-leaf ordering but $\hat{R}$ has a wider, α_Ω^* -scaled dynamic range — explaining why dips look deeper in the bottom row and why panels with large α_Ω^* (Jensen–Shannon, squared Euclidean) appear more stretched. The black dashed curve in each panel is that Ω ’s own π_Ω^* ; because all Ω are at their calibrated α_Ω^* , the seven π_Ω^* have comparable peak leaf-reach probabilities, so cross-panel comparison is fair at the geometric level. KL’s trained policy tracks π_{KL}^* within sampling noise across all 27 leaves; the six non-KL regularizers show systematic per-leaf error patterns that the scalar Δ_π integrates. The off-policy counterpart and the non-uniform π_{ref} panels are in Appendix B.17.

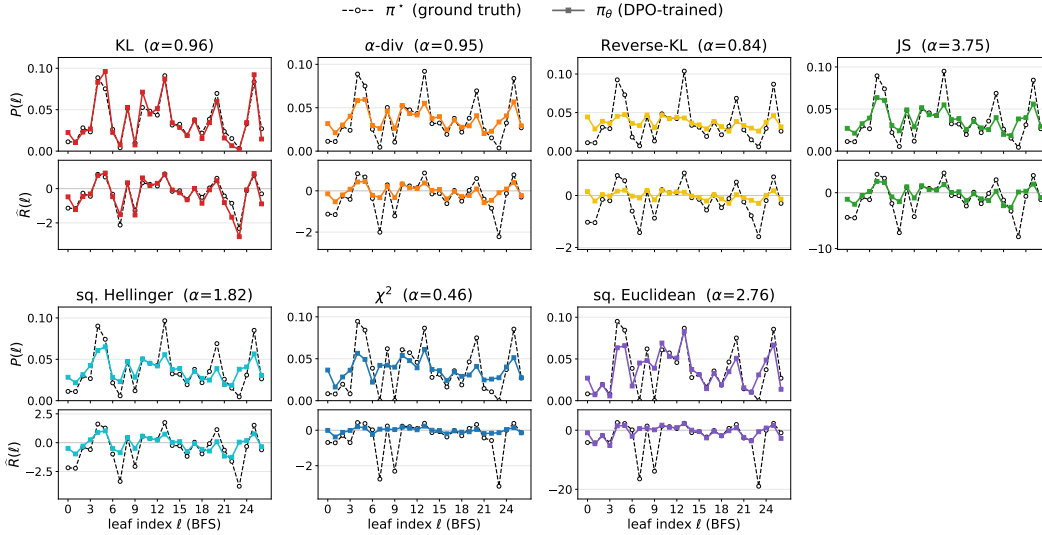


Figure 6: **Per-leaf anatomy of the trained policies**, on-policy, $H = 3$, calibrated α_Ω^* in each panel title. Top sub-panels: $P(\ell)$; bottom: $\hat{R}(\ell)$. Coloured solid = π_θ ; black dashed = that Ω ’s own π_Ω^* .

4.4 IS THE RECOVERY STABLE UNDER ON- VERSUS OFF-POLICY PREFERENCE DATA?

A policy trained from on-policy preferences (behavior = π_Ω^*) has the advantage of seeing exactly the state distribution it will be evaluated on. A policy trained from off-policy preferences (behavior = π_{ref} , the uniform reference) has to recover the same π_Ω^* from a behavior distribution that is generically different. For KL the inner term has zero variance under either behavior, so we predict the same Δ_π floor up to data noise; for the other Ω we predict a quantitative degradation, since the variance source is unchanged. The right panel of Figure 5 reads this prediction off the same experiment.

The off-policy curves match the on-policy curves up to small rearrangements: KL is again at $\Delta_\pi \approx 0.04$ with no n_{mc} trend, and the six non-KL regularizers trace the same $1/\sqrt{n_{\text{mc}}}$ -style approach to KL’s floor. Cross- Ω ordering is preserved across the regime switch. The recovery is stable: KL is essentially regime-invariant, and the relative ranking of the other regularizers carries over from on-policy to off-policy with no qualitative change. We verify in Appendix B.7 that the same conclusions hold under a non-uniform π_{ref} (peaked reference).

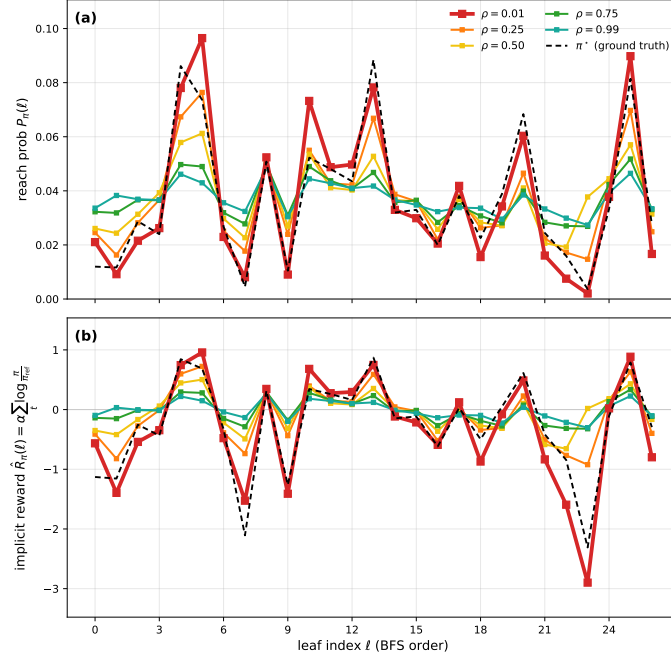


Figure 7: α -divergence parameter sweep, on-policy, $H = 3$. Top: per-leaf reach probability $P_{\pi_\theta}(\ell)$ at five values of ρ . Bottom: per-leaf DPO-implicit reward $\hat{R}_{\pi_\theta}(\ell)$. Black dashed: π^* and $\hat{R}^*$ at $\rho = 0.01$ (KL endpoint), shown once.

4.5 WHERE DOES THE GAP OPEN ALONG THE KL / REVERSE-KL FAMILY?

Q1–Q3 frame KL versus non-KL as a binary distinction. To see whether KL is a singular point or one end of a continuous boundary, we sweep the α -divergence family $\{f_\rho : \rho \in [0, 1]\}$ at $\rho \in \{0.01, 0.25, 0.50, 0.75, 0.99\}$, where $\rho \rightarrow 0$ recovers KL and $\rho \rightarrow 1$ recovers Reverse-KL through a Hellinger-like midpoint at $\rho = 0.5$. The preference dataset and ground-truth π^* are shared across ρ values so that any variation in the trained policy reflects the inner-term variance gap, not the underlying optimal policy.

Figure 7 shows the KL endpoint ($\rho = 0.01$, vivid red) tracking the dashed ground truth at every leaf. As ρ increases, the trained-policy curves fan out smoothly toward the Reverse-KL shape, with no special structure at any intermediate ρ . The gap is therefore a continuous boundary effect: off-policy admissibility is an endpoint property of the α -divergence family, not an isolated feature of $\rho = 0$. Off-policy counterparts of all main-text figures, the full $N_{\text{pairs}} \times n_{\text{mc}}$ heatmaps, the soft-target ECE diagnostic, the per-leaf anatomy of the Q2 trained policies, and the legacy compute-matched and vocabulary-scaling experiments are in Appendix B.

5 DISCUSSION

6 CONCLUSION

REPRODUCIBILITY STATEMENT

ETHICS STATEMENT

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A PROOFS FOR SECTION 3

We provide the proofs deferred from Section 3.

A.1 PROOF OF LEMMA 1 (STRUCTURAL FORM)

Proof. **Reduction to a concave program over occupancies.** For a policy π and initial state distribution ρ_0 , the (normalized) discounted state–action occupancy is

$$d(s, a) := (1 - \gamma) \sum_{t \geq 0} \gamma^t \mathbb{P}_\pi(s_t = s, a_t = a), \quad d(s) := \sum_a d(s, a).$$

The map sending π to its occupancy d is a bijection between policies and nonnegative measures on $\mathcal{S} \times \mathcal{A}$ satisfying the Bellman flow constraint

$$d(s') = (1 - \gamma) \rho_0(s') + \gamma \sum_{s, a} d(s, a) \mathbb{P}(s' | s, a) \quad \forall s' \in \mathcal{S},$$

where on the right-hand side (s, a) ranges over predecessor state–action pairs, with the policy recovered as $\pi(a|s) = \frac{d(s, a)}{d(s)}$ on reachable s . Under this change of variable, the Ω -regularized RL objective is, up to multiplication by $(1 - \gamma)^{-1}$,

$$\max_{d \geq 0} \sum_{s, a} d(s, a) r(s, a) - \alpha \sum_s d(s) \Omega\left(\frac{d(s, \cdot)}{d(s)}; \pi_{\text{ref}}(\cdot|s)\right),$$

subject to the Bellman flow constraint and $d \geq 0$. Each statewise summand $d(s) \Omega\left(\frac{d(s, \cdot)}{d(s)}; \pi_{\text{ref}}(\cdot|s)\right)$ is the perspective of the convex map $\Omega(\cdot; \pi_{\text{ref}}(\cdot|s))$ on $\Delta(\mathcal{A})$ and is therefore convex in $d(s, \cdot)$. The full objective is hence concave in d , the equality constraints are affine, and the nonnegativity constraints are convex, so the problem is a concave program; Slater’s condition holds at any strictly positive feasible d , and the KKT conditions are necessary and sufficient at the optimum.

Lagrangian. Introduce a multiplier $\lambda(s) \in \mathbb{R}$ for the Bellman flow constraint at each s , and a multiplier $\mu(s, a) \geq 0$ for the nonnegativity constraint $d(s, a) \geq 0$. The Lagrangian of the concave program is

$$\begin{aligned} \mathcal{L}(d, \lambda, \mu) &= \sum_{s, a} d(s, a) r(s, a) - \alpha \sum_s d(s) \Omega\left(\frac{d(s, \cdot)}{d(s)}; \pi_{\text{ref}}(\cdot|s)\right) \\ &\quad - \sum_{s'} \lambda(s') \left[d(s') - (1 - \gamma) \rho_0(s') - \gamma \sum_{s, a} d(s, a) \mathbb{P}(s' | s, a) \right] \\ &\quad + \sum_{s, a} \mu(s, a) d(s, a). \end{aligned}$$

At the optimum d^* we write $\pi^*(a|s) := d^*(s, a)/d^*(s)$ on reachable s , and the stationarity condition reads $\partial \mathcal{L} / \partial d(s, a) = 0$. We compute this derivative term by term below.

KKT stationarity. The contribution of $d(s, a)$ to each term of $\mathcal{L}$ is direct except for the regularizer term, which requires the chain rule with

$$\frac{\partial d(s)}{\partial d(s, a)} = 1 \quad \text{and} \quad \frac{\partial}{\partial d(s, a)} \left[\frac{d(s, b)}{d(s)} \right] = \frac{\mathbf{1}\{a = b\} - \pi^*(b|s)}{d^*(s)}.$$

Expanding the product rule and chain rule term by term gives the statewise perspective derivative:

$$\begin{aligned} &\frac{\partial}{\partial d(s, a)} \left[d(s) \Omega\left(\frac{d(s, \cdot)}{d(s)}; \pi_{\text{ref}}(\cdot|s)\right) \right]_{d=d^*} \\ &= \underbrace{\frac{\partial d(s)}{\partial d(s, a)}}_{=1} \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)) + d^*(s) \sum_b [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_b \cdot \frac{\partial}{\partial d(s, a)} \left[\frac{d(s, b)}{d(s)} \right]_{d=d^*} \\ &= \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)) + \sum_b [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_b (\mathbf{1}\{a = b\} - \pi^*(b|s)) \\ &= \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)) + [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a - \mathbb{E}_{b \sim \pi^*(\cdot|s)} [[\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_b] \\ &= [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a - \mathcal{C}_\Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)), \end{aligned}$$

where the last step uses the definition $C_\Omega(\pi; \pi_{\text{ref}}) := \mathbb{E}_{b \sim \pi}[\nabla_\pi \Omega(\pi; \pi_{\text{ref}})]_b - \Omega(\pi; \pi_{\text{ref}})$ from Equation 13 and the $d^*(s)$ factors cancel against the $1/d^*(s)$ in the chain-rule expression. Substituting this derivative into the stationarity condition $\partial \mathcal{L} / \partial d(s, a) = 0$ at $d = d^*$ yields

$$0 = r(s, a) - \alpha [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a + \alpha C_\Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)) - \lambda(s) + \gamma \sum_{s'} \mathbb{P}(s' | s, a) \lambda(s') + \mu(s, a).$$

Complementary slackness eliminates $\mu(s, a)$. The KKT complementary slackness condition reads $\mu(s, a) d^*(s, a) = 0$ for every (s, a) . The nonnegativity multiplier $\mu(s, a)$ can be positive only when the constraint $d(s, a) \geq 0$ is active, that is, when $d^*(s, a) = 0$. Essential smoothness of $\Omega(\cdot; \pi_{\text{ref}})$ in its first argument forces $|\nabla_\pi \Omega(\pi; \pi_{\text{ref}})| \rightarrow \infty$ as π approaches the relative boundary of $\Delta(\mathcal{A})$, so the per-state argmax of the regularized soft Bellman operator places $\pi^*(\cdot|s)$ strictly in the relative interior of $\Delta(\mathcal{A})$ at every state. Hence $\pi^*(a|s) > 0$ for every a , and $d^*(s, a) = d^*(s) \pi^*(a|s) > 0$ on every reachable (s, a) . Complementary slackness then forces $\mu(s, a) = 0$ on every reachable (s, a) . With $\mu \equiv 0$ the stationarity condition rearranges to

$$\lambda(s) = r(s, a) - \alpha [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a + \alpha C_\Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)) + \gamma \sum_{s'} \mathbb{P}(s' | s, a) \lambda(s'), \quad (21)$$

holding for every reachable (s, a) , with the multiplier λ not yet identified.

Identification of λ with V^* . Take the expectation of Equation 21 under $a \sim \pi^*(\cdot|s)$. The gradient term becomes

$$\alpha \mathbb{E}_{a \sim \pi^*(\cdot|s)} [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a = \alpha C_\Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)) + \alpha \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))$$

by the definition of C_Ω in Equation 13, and the αC_Ω contributions cancel. The recursion that remains is

$$\lambda(s) = \mathbb{E}_{a \sim \pi^*(\cdot|s)} [r(s, a)] - \alpha \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)) + \gamma \mathbb{E}_{a \sim \pi^*(\cdot|s), s' \sim \mathbb{P}(\cdot|s, a)} [\lambda(s')],$$

which is exactly the soft Bellman equation for the regularized value V^* evaluated under π^* . The soft Bellman operator at the policy π^* is a γ -contraction on bounded functions and has V^* as its unique fixed point, so $\lambda \equiv V^*$. Substituting $\lambda(s) = V^*(s)$ and $\lambda(s') = V^*(s')$ back into Equation 21 and recognizing $Q^*(s, a) := r(s, a) + \gamma \mathbb{E}_{s' \sim \mathbb{P}(\cdot|s, a)} [V^*(s')]$ gives

$$Q^*(s, a) = \alpha [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a + (V^*(s) - \alpha C_\Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))).$$

Defining

$$\beta(s) := V^*(s) - \alpha C_\Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)),$$

the stationarity condition reduces to $Q^*(s, a) = \alpha [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a + \beta(s)$, which is Equation 6. Substituting into the Bellman identity $r(s, a) = Q^*(s, a) - \gamma \mathbb{E}_{s' \sim \mathbb{P}(\cdot|s, a)} [V^*(s')]$ yields Equation 7. $\square$

A.2 PROOF OF THEOREM 1 (POLICY DEVIATION)

Proof. Define $\Delta(s) := V^*(s) - V^\pi(s)$. The soft Bellman identity for the regularized value reads, at any state s ,

$$\begin{aligned} V^*(s) &= \mathbb{E}_{a \sim \pi^*} [Q^*(s, a)] - \alpha \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)), \\ V^\pi(s) &= \mathbb{E}_{a \sim \pi} [Q^\pi(s, a)] - \alpha \Omega(\pi(\cdot|s); \pi_{\text{ref}}(\cdot|s)). \end{aligned}$$

Subtracting and inserting $\mathbb{E}_{a \sim \pi} [Q^*(s, a)]$ into the difference,

$$\Delta(s) = (\mathbb{E}_{a \sim \pi^*} [Q^*] - \mathbb{E}_{a \sim \pi} [Q^*]) + \mathbb{E}_{a \sim \pi} [Q^* - Q^\pi] - \alpha \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s)) + \alpha \Omega(\pi(\cdot|s); \pi_{\text{ref}}(\cdot|s)).$$

The Bellman identity $Q^*(s, a) - Q^\pi(s, a) = \gamma \mathbb{E}_{s' \sim \mathbb{P}(\cdot|s, a)} [\Delta(s')]$ rewrites the middle term as $\gamma \mathbb{E}_{a \sim \pi, s' \sim \mathbb{P}} [\Delta(s')]$. For the first bracket, Lemma 1 gives $Q^*(s, a) = \alpha [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a + \beta(s)$. The $\beta(s)$ contribution cancels under $\mathbb{E}_{\pi^*} - \mathbb{E}_\pi$, leaving

$$\mathbb{E}_{a \sim \pi^*} [Q^*] - \mathbb{E}_{a \sim \pi} [Q^*] = \alpha (\mathbb{E}_{a \sim \pi^*} [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a - \mathbb{E}_{a \sim \pi} [\nabla_\pi \Omega(\pi^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a).$$

Combining with the two Ω terms and recognizing Equation 8,

$$\Delta(s) = \alpha B_\Omega(\pi(\cdot|s) \parallel \pi^*(\cdot|s)) + \gamma \mathbb{E}_{a \sim \pi, s' \sim \mathbb{P}}[\Delta(s')].$$

Unrolling this recursion under π starting at $s_0 = s$ gives

$$\Delta(s) = \alpha \mathbb{E}_{\tau \sim \mathbb{P}_\pi} \left[\sum_{l \geq 0} \gamma^l B_\Omega(\pi(\cdot|s_l) \parallel \pi^*(\cdot|s_l)) \right],$$

where $\tau = (s_0, s_1, s_2, \dots)$ denotes a trajectory generated under π from initial state $s_0 = s$.

To convert this state-value recursion into the action-value form of Equation 9, apply the Bellman identity $Q^*(s, a) - Q^\pi(s, a) = \gamma \mathbb{E}_{s' \sim \mathbb{P}(\cdot|s, a)}[\Delta(s')]$ and substitute the unrolled expression for $\Delta(s')$. A trajectory rooted at (s, a) , written $\tilde{\tau} = (s, a, s'_1, s'_2, \dots) \sim \mathbb{P}_{s, a}^\pi$, satisfies $s'_1 \sim \mathbb{P}(\cdot|s, a)$ and $a'_t \sim \pi(\cdot|s'_t)$ thereafter. The unrolled trajectory used in $\Delta(s')$ then starts at $s_0 = s' = s'_1$, so its l -th visited state s_l corresponds to s'_{l+1} along $\tilde{\tau}$. Collecting the discount factors,

$$\begin{aligned} Q^*(s, a) - Q^\pi(s, a) &= \gamma \cdot \alpha \mathbb{E}_{\tilde{\tau}} \left[\sum_{l \geq 0} \gamma^l B_\Omega(\pi(\cdot|s'_{l+1}) \parallel \pi^*(\cdot|s'_{l+1})) \right] \\ &= \alpha \mathbb{E}_{\tilde{\tau}} \left[\sum_{k \geq 1} \gamma^k B_\Omega(\pi(\cdot|s'_k) \parallel \pi^*(\cdot|s'_k)) \right] = \alpha \bar{B}_\Omega(\pi \parallel \pi^*; s, a), \end{aligned}$$

where the second line uses the change of index $k = l + 1$. The final expression matches Equation 10, completing the proof of Equation 9. $\square$

A.3 PROOF OF THEOREM 2 (KL UNIQUENESS)

Proof. We work pointwise at a fixed next state $s' \in \mathcal{S}$, with $\pi := \pi(\cdot|s')$, $\pi_{\text{ref}} := \pi_{\text{ref}}(\cdot|s')$, and the action index a ranging over $\mathcal{A}$ (equivalently the next-state action a'). Axiom (A4) is interpreted in this pointwise form: at the fixed s' , the functional $C_\Omega(\pi(\cdot|s'); \pi_{\text{ref}}(\cdot|s'))$ is independent of $\pi(\cdot|s')$. The per-action functions g_a and the constants $c, d_a, e_a, l_a, \lambda$ introduced below may depend on $\pi_{\text{ref}}(\cdot|s')$ and hence implicitly on s' ; we suppress this dependence in the notation and reinstate it at the end of the proof.

Step 1. ODE from axioms (A4) and (A5). By Axiom (A5) we may write $\Omega(\pi; \pi_{\text{ref}}) = \sum_a g_a(\pi(a))$, hence $[\nabla_\pi \Omega(\pi; \pi_{\text{ref}})]_a = g'_a(\pi(a))$. The inner-term functional in Equation 13 becomes

$$C_\Omega(\pi; \pi_{\text{ref}}) = \sum_a [\pi(a) g'_a(\pi(a)) - g_a(\pi(a))] = \sum_a h_a(\pi(a)),$$

where $h_a(t) := t g'_a(t) - g_a(t)$. Consider any simplex-tangent perturbation $\pi \mapsto \pi + \epsilon(e_i - e_j)$, which preserves $\sum_a \pi(a) = 1$. Axiom (A4) requires $dC_\Omega/d\epsilon|_{\epsilon=0} = 0$, that is,

$$h'_i(\pi(i)) - h'_j(\pi(j)) = 0 \quad \forall i \neq j, \forall \pi \in \Delta(\mathcal{A}).$$

Since $\pi(i)$ and $\pi(j)$ vary independently in $(0, 1)$, this forces $h'_i(t) = c$ for some constant c shared across all i . Equivalently, $h_i(t) = ct + l_i$ for action-dependent constants l_i .

Step 2. Two integrations. The relation $t g'_i(t) - g_i(t) = ct + l_i$ differentiates in t to

$$g'_i(t) + t g''_i(t) - g'_i(t) = c,$$

that is, $g''_i(t) = c/t$. Integrating once gives $g'_i(t) = c \log t + d_i$ for some constant d_i , and integrating once more gives

$$g_i(t) = c(t \log t - t) + d_i t + e_i,$$

for some constant e_i . A direct check confirms $g'_i(t) = c \log t + d_i$ from this expression. Strict convexity from Axiom (A3) requires $g''_i(t) = c/t > 0$ for all $t > 0$, so $c > 0$.

Step 3. Pin d_a via the minimum at the reference. Axiom (A2) makes $\pi = \pi_{\text{ref}}$ the unique global minimum of $\Omega(\cdot; \pi_{\text{ref}})$ on $\Delta(\mathcal{A})$. Simplex stationarity at this minimum reads $\nabla_\pi \Omega(\pi_{\text{ref}}; \pi_{\text{ref}}) = \lambda \mathbf{1}$

for some scalar λ . Using $[\nabla_{\pi}\Omega(\pi; \pi_{\text{ref}})]_a = g'_a(\pi(a)) = c \log \pi(a) + d_a$ from Step 2 and evaluating at $\pi = \pi_{\text{ref}}$,

$$c \log \pi_{\text{ref}}(a) + d_a = \lambda \quad \forall a,$$

so $d_a = \lambda - c \log \pi_{\text{ref}}(a)$.

Step 4. Assemble and apply Axiom (A1). Substituting the pinned d_a into Step 2 and summing over actions,

$$\begin{aligned} \Omega(\pi; \pi_{\text{ref}}) &= \sum_a g_a(\pi(a)) \\ &= \sum_a \left[c \pi(a) \log \pi(a) - c \pi(a) + (\lambda - c \log \pi_{\text{ref}}(a)) \pi(a) + e_a \right] \\ &= c \sum_a \pi(a) \log \pi(a) - c \sum_a \pi(a) \log \pi_{\text{ref}}(a) + (\lambda - c) \sum_a \pi(a) + \sum_a e_a \\ &= c D_{\text{KL}}(\pi \| \pi_{\text{ref}}) + [\lambda - c + \sum_a e_a], \end{aligned}$$

using $\sum_a \pi(a) = 1$ on the simplex. Axiom (A1) at $\pi = \pi_{\text{ref}}$ gives $0 = 0 + [\lambda - c + \sum_a e_a]$, so the bracketed constant vanishes. Hence

$$\Omega(\pi; \pi_{\text{ref}}) = c D_{\text{KL}}(\pi \| \pi_{\text{ref}}), \quad c = c(\pi_{\text{ref}}) > 0,$$

which is Equation 17. The converse, that $c D_{\text{KL}}$ satisfies axioms (A1) through (A5) with $C_{\Omega} \equiv c$, is a direct verification. Reinstating the suppressed s' -dependence, the conclusion reads

$$\Omega(\pi(\cdot|s'); \pi_{\text{ref}}(\cdot|s')) = c(s') \cdot D_{\text{KL}}(\pi(\cdot|s') \| \pi_{\text{ref}}(\cdot|s')),$$

where $c(s') := c(\pi_{\text{ref}}(\cdot|s')) > 0$. The scalar may vary across next states s' insofar as the reference policy does; whenever π_{ref} is shared across states (e.g. uniform), c reduces to a single constant. $\square$

A.4 f -DIVERGENCE REDUCTION OF THEOREM 2

The inline argument in Section 3.4 deferred the explicit ODE manipulation. We record it here for completeness. For the f -divergence parameterization $\Omega(\pi; \pi_{\text{ref}}) = D_f(\pi \| \pi_{\text{ref}})$ defined in Equation 1, axioms (A1)–(A3) follow from the standard properties of f -divergences (convex f with $f(1) = 0$). Axiom (A5) holds with $g_a(t) = \pi_{\text{ref}}(a) f(t/\pi_{\text{ref}}(a))$. For Axiom (A4), the inner-term functional in Equation 13 evaluates to

$$C_{\Omega}(\pi; \pi_{\text{ref}}) = \mathbb{E}_{a \sim \pi} \left[f' \left(\frac{\pi(a)}{\pi_{\text{ref}}(a)} \right) \right] - \mathbb{E}_{a \sim \pi_{\text{ref}}} \left[f \left(\frac{\pi(a)}{\pi_{\text{ref}}(a)} \right) \right] = \mathbb{E}_{a \sim \pi} \left[\Phi_f \left(\frac{\pi(a)}{\pi_{\text{ref}}(a)} \right) \right],$$

where the second equality rewrites $\mathbb{E}_{a \sim \pi_{\text{ref}}} [f(\pi(a)/\pi_{\text{ref}}(a))] = \mathbb{E}_{a \sim \pi} [f(\pi(a)/\pi_{\text{ref}}(a)) \pi_{\text{ref}}(a)/\pi(a)]$ by importance sampling, and uses the definition $\Phi_f(t) := f'(t) - f(t)/t$. For C_{Ω} to be constant in π , $\Phi_f(t)$ must be constant in t . Otherwise the expectation depends on the support of π/π_{ref} . The condition $\Phi_f(t) \equiv c$ rearranges to $t f'(t) - f(t) = ct$, which differentiates to $t f''(t) = c$ and integrates twice to $f(t) = c(t \log t - t) + dt + e$. The condition $f(1) = 0$ fixes $e = c - d$, so $f(t) = ct \log t + (d - c)(t - 1)$. The second term is exactly the f -divergence equivalence freedom $f(t) \mapsto f(t) + \kappa(t - 1)$ that leaves D_f invariant, since $\sum_a \pi_{\text{ref}}(a) (\pi(a)/\pi_{\text{ref}}(a) - 1) = 0$ on the simplex. Hence $D_f = c \cdot D_{\text{KL}}$ uniquely, recovering Theorem 2 within the f -divergence parameterization.

A.5 TERMINOLOGY: SEPARABILITY AND DECOMPOSABILITY

Axiom (A5) uses *separability* in the standard convex-analysis sense (Boyd & Vandenberghe, 2004, §5): the functional decomposes as a sum of single-coordinate components, $\Omega(\pi; \pi_{\text{ref}}) = \sum_a g_a(\pi(a))$. The same property appears under other names. In Amari's information geometry it is called *decomposability* (Amari, 2016, Chapter 2), and every Csiszár f -divergence (Csiszár, 1967) is decomposable in this sense by construction. We keep the convex-analysis term because the regularized-MDP analysis here leans most directly on that literature.

A.6 DISCUSSION: DROPPING OR WEAKENING AXIOM (A5)

A natural follow-up question is what survives when Axiom (A5) is dropped, or weakened to allow a non-additive per-action decomposition. We collect what the remaining axioms still imply, and check the standard non-separable candidates.

What survives without (A5). Step 1 of the proof of Theorem 2 (Appendix A) uses (A5) to write $[\nabla_\pi \Omega(\pi; \pi_{\text{ref}})]_a = g'_a(\pi(a))$, after which Axiom (A4) reduces to the per-action ODE $h'_a(\pi(a)) = c$ via the simplex-tangent perturbation $\pi \mapsto \pi + \epsilon(e_i - e_j)$. Without (A5), $[\nabla_\pi \Omega]_a$ generically depends on the full vector π , and Axiom (A4) becomes a partial differential equation on C_Ω over the simplex: the same simplex-tangent argument gives only

$$\frac{\partial C_\Omega}{\partial \pi(i)} - \frac{\partial C_\Omega}{\partial \pi(j)} = 0 \quad \forall i \neq j, \forall \pi \in \Delta(\mathcal{A}),$$

which does not decouple into a single-coordinate ODE.

Standard non-separable candidates fail (A4). The most natural strictly convex non-separable functionals on the simplex all violate Axiom (A4):

- *Quadratic forms* $\Omega(\pi) = \frac{1}{2} \pi^\top M \pi$ with M positive definite. Then $[\nabla \Omega]_a = (M\pi)_a$ and $C_\Omega(\pi) = \pi^\top M \pi - \frac{1}{2} \pi^\top M \pi = \frac{1}{2} \pi^\top M \pi$, which is constant in π only when $M = 0$.
- *Power norms* $\Omega(\pi) = \frac{1}{p} \|\pi\|_p^p$ with $p > 1$. Then $[\nabla \Omega]_a = \pi(a)^{p-1}$ and $C_\Omega(\pi) = (1 - 1/p) \|\pi\|_p^p$, again non-constant.
- *Monotone outer transforms* $\Omega(\pi) = h(\sum_a g(\pi(a)))$ with strictly increasing convex h and strictly convex g . Writing $\Sigma(\pi) := \sum_a g(\pi(a))$, $C_\Omega(\pi) = h'(\Sigma) \sum_a \pi(a) g'(\pi(a)) - h(\Sigma)$. As π varies, Σ varies, so this is constant in π only when h is affine, which collapses the construction back to the separable case $\sum_a g(\pi(a))$ up to constants.

Separable inputs with a non-additive aggregator collapse to (A5). A more refined question is whether weakening (A5) to “each coordinate of Ω depends on a single $\pi(a)$, but the aggregator is non-additive” admits new examples. Concretely, take

$$\Omega(\pi) = h(g_1(\pi(a_1)), g_2(\pi(a_2)), \dots, g_n(\pi(a_n)))$$

for a strictly increasing convex $h : \mathbb{R}^n \rightarrow \mathbb{R}$ and per-coordinate strictly convex $g_a : (0, 1) \rightarrow \mathbb{R}$. Writing $u_a := g_a(\pi(a))$ and $h_a := \partial h / \partial u_a$, $h_{ab} := \partial^2 h / \partial u_a \partial u_b$, the chain rule gives $[\nabla_\pi \Omega]_a = h_a(u) g'_a(\pi(a))$, hence

$$C_\Omega(\pi) = \sum_a \pi(a) h_a(u) g'_a(\pi(a)) - h(u).$$

Applying the simplex-tangent equality $\partial_i C_\Omega = \partial_j C_\Omega$ and using $\partial u_b / \partial \pi(i) = g'_i(\pi(i)) \mathbf{1}\{b = i\}$, a direct computation gives

$$\partial_i C_\Omega = g'_i(\pi(i)) \sum_a \pi(a) g'_a(\pi(a)) h_{ai}(u) + \pi(i) h_i(u) g''_i(\pi(i)).$$

For $\partial_i C_\Omega = \partial_j C_\Omega$ to hold for every interior π and every pair $i \neq j$, the off-diagonal cross-partials h_{ai} must vanish identically: any nonzero cross-partial introduces a $\pi(a)$ -dependence on one side that the other side cannot match without coincidentally identical g'_a , g_a , and $\pi(a)$ entries. With $h_{ai} \equiv 0$ for all $a \neq i$, the first partial $h_i := \partial h / \partial u_i$ depends on u_i alone (its derivatives in every other coordinate vanish), so $h_i(u) = H_i(u_i)$ for some single-variable function H_i . Choosing an antiderivative $h_i(t) := \int H_i(t) dt$ for each i gives a candidate separable function $\tilde{h}(u) := \sum_a h_a(u_a)$ whose gradient matches that of h pointwise. Hence $h - \tilde{h}$ has zero gradient on the open domain and is constant, so $h(u) = \sum_a h_a(u_a)$ up to an additive constant (which contributes a constant shift to Ω and is absorbed into the shaping freedom). Substituting back gives $\Omega(\pi) = \sum_a h_a(g_a(\pi(a)))$, which is precisely the additive separable form (A5) with new per-coordinate functions $\tilde{g}_a := h_a \circ g_a$. The non-additive aggregator therefore collapses to the additive case, and Theorem 2 applies as before.

Conjecture. The pattern above, combined with the proof of Theorem 2 under (A5), leads us to conjecture that no strictly convex genuinely non-separable divergence on $\Delta(\mathcal{A})$ satisfies Axioms (A1)–(A4) simultaneously. A proof for the fully non-separable case (where each $[\nabla_\pi \Omega]_a$ depends on multiple $\pi(b)$ coordinates) or an explicit counterexample would close the only remaining axiomatic gap in the KL uniqueness characterization.

B ADDITIONAL EXPERIMENTAL DETAILS

This appendix collects the supporting material for Section 4. Appendix B.1 specifies the toy tree MDP environment. Appendix B.2 gives the closed-form f , f' , and Φ_f for the four divergence baselines. Appendix B.3 derives the trajectory-DPO loss for a general Ω . Appendix B.4 is the hyperparameter table. Appendix B.8 and Appendix B.9 report two diagnostic experiments that isolate the Monte-Carlo variance of the inner term: a single-state estimator-variance sweep over the per-call sample budget n , and a trajectory-level compounding sweep over the rollout horizon H with no training. Appendix B.10 reports a complementary view of the compute-matched experiment, where N_{steps} is held constant and n_{mc} is swept on its own. Appendix B.11 discusses the depth panel of the original training sweep, in which all regularizers (including KL) incur a budget-per-state confound that the compute-matched view in Appendix B.15 avoids. Source code is in `experiments/01_inner_term_variance.py`, `experiments/02_trajectory_mc.py`, `experiments/02_dpo_sweeps.py`, `experiments/03_compute_matched.py`, and `experiments/04_vocab_scaling.py`, with shared utilities in `divergences.py`, `tree_mdp.py`, and `dpo.py`.

B.1 TOY TREE MDP

The multi-step environment is a complete tree with width $W = 3$ and horizon H , indexed in BFS order. The root is state 0; node i 's children under actions $a \in \{0, 1, 2\}$ are $W \cdot i + 1 + a$. There are $|\mathcal{S}| = (W^{H+1} - 1)/(W - 1)$ states, of which W^H are leaves. Transitions are deterministic. Per-state rewards are 0 at non-leaves and $r(\ell) \sim \mathcal{N}(0, 1)$ at leaves (sampled once per random seed and held fixed for that seed). Trajectories run from the root for exactly H steps, ending at a leaf. The reference policy is uniform: $\pi_{\text{ref}}(a|s) = 1/W$ at every non-leaf s .

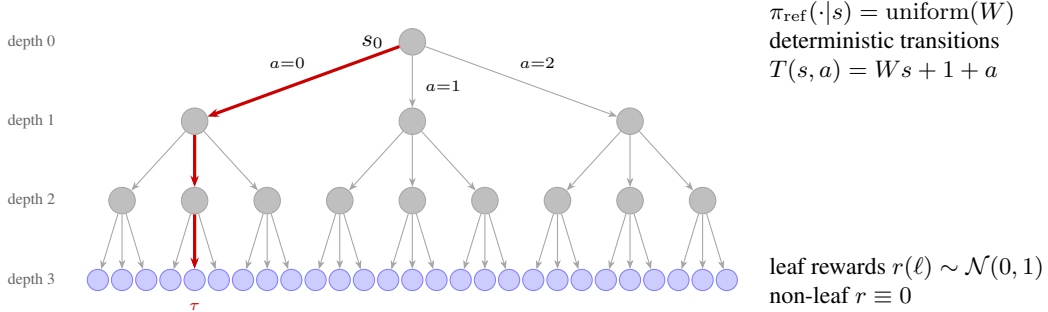


Figure 8: **Toy tree MDP topology** ($W = 3$, $H = 3$, $|\mathcal{S}| = 40$). Internal nodes (gray) are non-leaf states. Leaves (blue) carry i.i.d. rewards $r(\ell) \sim \mathcal{N}(0, 1)$. From every non-leaf state, three actions $a \in \{0, 1, 2\}$ deterministically reach the indexed children. The reference policy π_{ref} is uniform. The bold red path shows one trajectory $\tau = (s_0, a_0 = 0, s_1, a_1 = 1, s_2, a_2 = 1, s_3 = \ell_4)$ sampled in trajectory-DPO training. Preferences are generated by a Bradley–Terry oracle on pairs of such trajectories.

B.2 DIVERGENCE DEFINITIONS

For an f -divergence $\Omega(\pi; \pi_{\text{ref}}) = \sum_a \pi_{\text{ref}}(a) f(\pi(a)/\pi_{\text{ref}}(a))$ the action-wise gradient is $\nabla \Omega(\pi)(a) = f'(\pi(a)/\pi_{\text{ref}}(a))$ and the off-policyness obstruction $\Phi_f(t) := f'(t) - f(t)/t$ controls whether the inner term $C_\Omega(\pi) = \mathbb{E}_{a \sim \pi} [\Phi_f(\pi/\pi_{\text{ref}})]$ reduces to a constant. Closed forms for the four baselines:

B.3 TRAJECTORY-DPO LOSS FOR GENERAL Ω

For a general f -divergence regularizer Ω , the implicit reward representation evaluated at a transition (s, a, s') is, up to a state-dependent potential,

$$\hat{r}_\theta(s, a) = \alpha \left[\nabla \Omega(\pi_\theta(\cdot|s))(a) - \nabla \Omega(\pi_{\text{ref}}(\cdot|s))(a) \right] + \alpha C_\Omega(\pi_\theta(\cdot|s')). \quad (22)$$

Divergence	$f(t)$	$f'(t)$	$\Phi_f(t)$	$C_\Omega(\pi)$ at uniform π_{ref}
KL	$t \log t$	$\log t + 1$	$\equiv 1$	1 (constant)
Reverse-KL	$-\log t$	$-1/t$	$(\log t - 1)/t$	π -dependent
χ^2	$(t - 1)^2$	$2(t - 1)$	$2(t - 1) - (t - 1)^2/t$	π -dependent
squared Hellinger	$(\sqrt{t} - 1)^2$	$1 - 1/\sqrt{t}$	$(1 - 1/\sqrt{t}) - (\sqrt{t} - 1)^2/t$	π -dependent

Table 2: Closed-form f , f' , and Φ_f for the four f -divergence baselines. KL is the unique row with Φ identically constant.

For each preference pair (τ_w, τ_l) , the trajectory-DPO loss aggregates this implicit reward across all transitions of the trajectory:

$$\mathcal{L}_\Omega = -\log \sigma \left(\beta_{\text{DPO}} \left[\sum_{t=0}^{T-1} \hat{r}_\theta(s_t^w, a_t^w) - \sum_{t=0}^{T-1} \hat{r}_\theta(s_t^l, a_t^l) \right] \right). \quad (23)$$

When s_{t+1} is a terminal leaf the inner term $C_\Omega(\pi_\theta(\cdot|s_{t+1}))$ does not appear (no actions to expectation-over). For non-terminal s_{t+1} and non-KL Ω , we estimate C_Ω by Monte Carlo with n_{mc} samples (Option A in our framework). For KL specifically, $C_{\text{KL}} \equiv 1$ is a state-independent constant that cancels in the $\hat{r}_\theta(s, a_w) - \hat{r}_\theta(s, a_l)$ pairing, recovering standard DPO with implicit reward $\beta \log(\pi_\theta/\pi_{\text{ref}})$ modulo a trajectory-length-dependent constant.

B.4 HYPERPARAMETERS

Symbol	Value	Description
W	3	Tree width (number of actions per non-leaf state)
H	3 (default)	Trajectory horizon = tree depth (varied in App. B.9, App. B.11)
α	1.0	Ω regularization weight
β_{or}	1.0	Bradley–Terry oracle temperature
β_{DPO}	1.0	DPO loss temperature
N_{pairs}	1000	Preference dataset size per training run
N_{steps}	500	Adam steps per training run
batch size	64	Preference pairs per gradient update
learning rate	0.05	Adam learning rate on tabular logits
n_{mc}	8 (default)	MC samples per non-terminal next state (varied in Sec. B.10)
N_{seeds}	50 (single config), 20 (sweeps)	Independent seeds for std estimation

Table 3: Hyperparameters for the multi-step trajectory-DPO experiments (Section B.10 and Appendix B.9). The single-state inner-term variance experiment (Appendix B.8) uses $|\mathcal{A}| = 100$, peakedness 3.0, 500 trials per n , sweeping $n \in \{2^2, 2^3, \dots, 2^{10}\}$.

B.5 PER- Ω REGULARIZATION-WEIGHT CALIBRATION

At a shared $\alpha = 1$, each Ω ’s soft-Bellman optimal policy π_Ω^* has a different peak leaf-reach probability $\max_\ell P_{\pi_\Omega^*}(\ell)$ because each Ω has a distinct geometric scale. To remove this artefact from cross- Ω comparisons we calibrate α_Ω^* per Ω so all π_Ω^* have the same peak.

Anchor. KL at $\alpha = 1$, peak target $\bar{p} := \mathbb{E}_{\text{seed}}[\max_\ell P_{\pi_{\text{KL}}^*}(\ell)] \approx 0.19$ over 5 leaf-reward draws.

Procedure. For each non-KL Ω , sweep α on a log grid of 25 values in $[0.05, 5.0]$. At each α , compute π_Ω^* by backward induction and record the peak leaf-reach probability. Interpolate in $\log \alpha$ to find the unique crossing α_Ω^* where the peak matches $\bar{p}$. Repeat for 5 leaf-reward seeds; report the median as the main-text α_Ω^* and the min/max as the range. The calibration runs on closed-form backward induction only and completes in seconds.

Source: `experiments/22_alpha_peak_match.py`.

Table 4: **Calibrated** α_Ω^* . Median across 5 leaf-reward seeds, peak target $\bar{p} \approx 0.19$. Min/max in parentheses give the per-seed range.

Ω	median α_Ω^*	min / max
KL	0.96	0.37/1.51
α -divergence ($\alpha_{\text{div}} = 0.1$)	0.95	0.37/1.50
Reverse-KL	0.84	0.36/1.40
Jensen-Shannon	3.75	1.51/5.00
squared Hellinger	1.82	0.74/2.92
Pearson χ^2	0.46	0.16/0.74
squared Euclidean	2.76	0.97/4.48

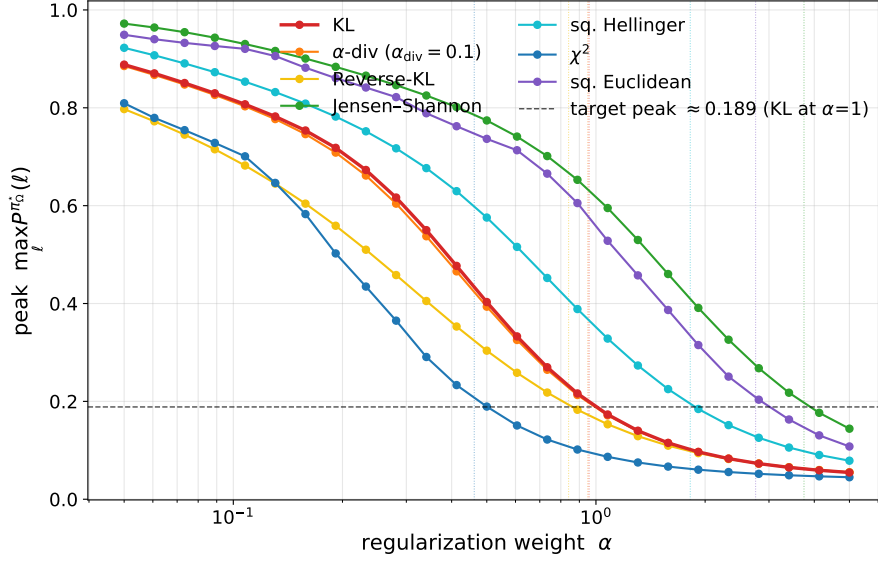


Figure 9: **Peak leaf-reach probability of π_Ω^* vs α , median over 5 seeds.** Dashed horizontal: KL anchor $\bar{p} \approx 0.19$. Dotted verticals: α_Ω^* per Table 4.

B.6 REWARD-SPACE COMPANION METRIC Δ_r AND UNCALIBRATED BASELINE

The main text uses the policy-space TV gap Δ_π (Equation 20) because it is bounded in $[0, 1]$ for every Ω and free of regularizer scale. We document here the reward-space companion that motivated earlier drafts.

For each Ω , the *canonical reward* that induces π_Ω^* under the shaping choices $\alpha = 1$ and $\beta(s) \equiv 0$ of Lemma 1 is

$$r_{\Omega, \text{eq}}^*(s, a) := \alpha [\nabla_{\pi_\Omega} \Omega(\pi_\Omega^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))]_a - \gamma V_{\Omega, \text{eq}}^*(\text{child}(s, a)), \quad (24)$$

with $V_{\Omega, \text{eq}}^*(s) := \alpha C_\Omega(\pi_\Omega^*(\cdot|s); \pi_{\text{ref}}(\cdot|s))$ at non-leaves and $V_{\Omega, \text{eq}}^*(\text{leaf}) := 0$. The trained policy π_θ yields an empirical implicit reward $\hat{r}_\theta$ via the same equation with π_θ in place of π_Ω^* , and the equivalent-reward gap is

$$\Delta_r(\Omega) := \mathbb{E}_{(s, a) \sim d^{\pi_\Omega^*}} [\hat{r}_\theta(s, a) - r_{\Omega, \text{eq}}^*(s, a)]. \quad (25)$$

Figure 10 shows the uncalibrated baseline ($\alpha = 1$ for every Ω , Δ_r metric) — the original Figure 5 from earlier drafts. The main qualitative conclusion (KL flat at floor, the four Φ -bounded Ω converging to it) is preserved, but two cosmetic issues become caveats: (i) Pearson χ^2 's Δ_r stays small because at $\alpha = 1$ the χ^2 penalty pulls $\pi_{\chi^2}^*$ close to the uniform π_{ref} , so the gap measures a near-trivial recovery; (ii) Jensen-Shannon and squared Euclidean's Δ_r are larger because their reward-space gradient scale at $\alpha = 1$ amplifies any residual policy mismatch. The calibrated Δ_π in the main text removes both artefacts.

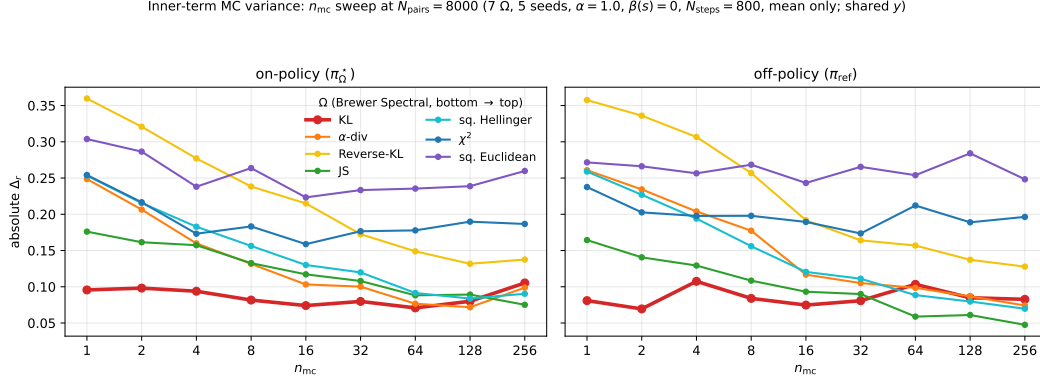


Figure 10: **Reward-space gap Δ_r at $\alpha = 1$, uncalibrated baseline.** Companion to Figure 5; mean across 5 seeds.

B.7 ROBUSTNESS ABLATIONS: PEAKED π_{ref} AND α -DIVERGENCE FAMILY SWEEP

The main-text Δ_π result (Figure 5) uses uniform π_{ref} and treats KL as an isolated point against six discrete Ω . We document two robustness ablations.

Non-uniform π_{ref} . We repeat the Figure 5 experiment with a fixed peaked reference policy ($\pi_{ref} = \text{softmax of } \mathcal{N}(0, 1) \text{ logits, } \min_a \pi_{ref}(a|s) = 0.01, \max = 0.87$). All seven Ω use their calibrated α_Ω^* from Table 4. Figure 11 shows that the qualitative story (KL lowest-and-flat, six non-KL converging) is preserved: KL stays at $\Delta_\pi \approx 0.04$ across n_{mc} , and the relative ordering of non-KL Ω is unchanged.

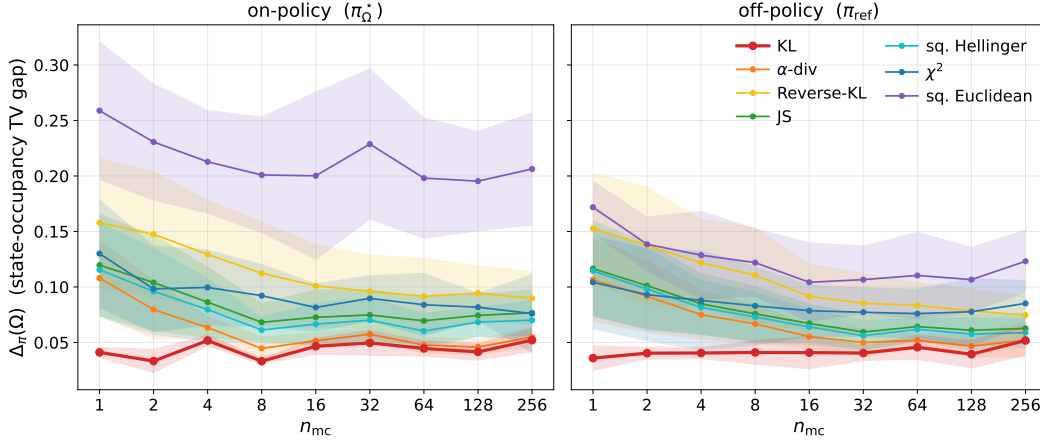


Figure 11: **Non-uniform π_{ref} counterpart of Figure 5, peakedness 1.0, seed 999.**

α -divergence family sweep. Four of our seven Ω are members of the α -divergence family at specific parameter values: KL ($\alpha_{div} \rightarrow 0$), Reverse-KL ($\alpha_{div} \rightarrow 1$), squared Hellinger ($\alpha_{div} = 0.5$, up to a factor 2), and Pearson χ^2 ($\alpha_{div} = -1$, up to a factor $1/2$). To test whether KL is an isolated point or one end of a continuous boundary, we sweep $\alpha_{div} \in \{-1, -0.5, 0, 0.5, 1\}$ at $\alpha = 1$ (no calibration in this ablation) and report Δ_π in Figure 12. KL ($\alpha_{div} = 0$) is the unique minimum-and-flat point of the family; moving away in either direction raises the floor monotonically.

B.8 SINGLE-STATE INNER-TERM VARIANCE: NUMERIC SUPPLEMENT TO SECTION 4.2

Table 5 reports concrete standard-deviation values at three representative budgets for the experiment plotted in Figure 3. The KL row is identically zero because $C_{KL} \equiv 1$, so every n -sample average

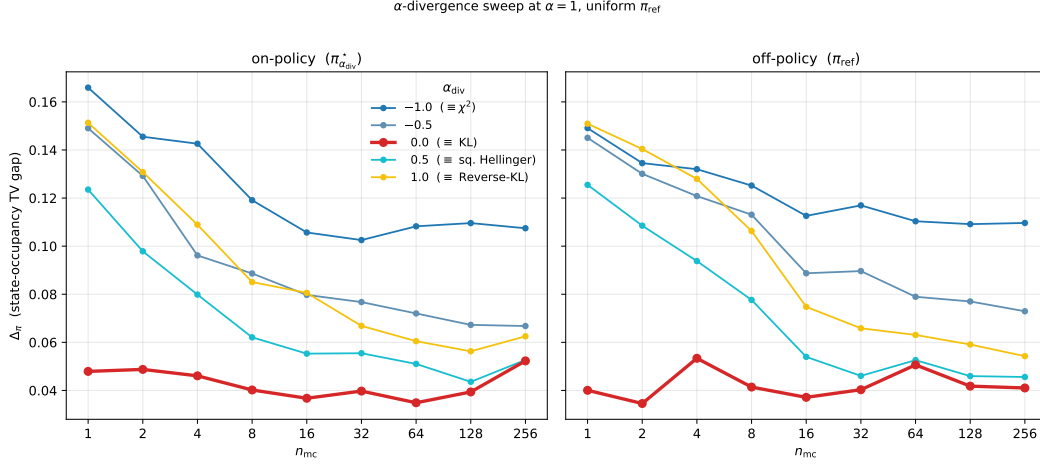


Figure 12: α -divergence family sweep. Δ_π vs n_{mc} at $\alpha_{div} \in \{-1, -0.5, 0, 0.5, 1\}$; endpoints use closed-form KL/Reverse-KL.

returns exactly the same value. The other three rows decay as $1/\sqrt{n}$ as predicted by the central limit theorem, with the constant set by the shape of the inner-term integrand on the support of π/π_{ref} . Reverse-KL has the largest constant because of the $1/t$ blow-up of its integrand on low-probability actions; χ^2 and squared Hellinger are intermediate.

Table 5: **Single-state inner-term variance, numeric values.** Standard deviation of the n -sample Monte-Carlo estimator of $C_\Omega(\pi; \pi_{ref})$ over 500 independent runs at $|\mathcal{A}| = 100$ and peakedness 3.0. KL is the unique divergence with identically zero estimator standard deviation.

Divergence Ω	std at $n = 4$	std at $n = 64$	std at $n = 1024$
KL	0.000	0.000	0.000
Reverse-KL	26.35	11.51	5.42
χ^2	6.39	2.03	0.70
squared Hellinger	3.74	1.53	0.61

B.9 TRAJECTORY MONTE-CARLO COMPOUNDING: NUMERIC SUPPLEMENT TO SECTION 4.2

Table 6 reports concrete per-trajectory standard-deviation values at four representative horizons for the experiment plotted in Figure 4. The KL row is identically zero at every H to floating-point precision, the most direct empirical confirmation of off-policy admissibility. The non-KL rows show the predicted $\sqrt{H-1}$ growth, with mild over-shooting due to varying per-state inner-term variance across visited next-states. At $H = 8$ the Reverse-KL trajectory standard deviation reaches 2.81 while squared Euclidean stays near 0.08.

B.10 n_{mc} SWEEP AT FIXED N_{steps} (COMPLEMENT TO APPENDIX B.15)

The compute-matched view in Appendix B.15 charges each non-KL regularizer for the cost of its Monte-Carlo inner term. A complementary view holds the number of gradient steps N_{steps} constant and sweeps n_{mc} . Under this convention KL is invariant in n_{mc} because C_{KL} is a constant that drops out of the chosen-minus-rejected pairing, and non-KL improves with larger n_{mc} as the CLT $1/\sqrt{n_{mc}}$ rate predicts.

Table 7 reads non-KL improvement with larger n_{mc} as evidence that the Monte-Carlo variance does penalize trained-policy quality. The compute-matched view in Appendix B.15 answers the natural follow-up: at fixed total compute, does larger n_{mc} pay for itself, given that it reduces N_{steps} ? The answer in our setting is that it does for non-KL up to $n_{mc} \approx 16$, after which the optimization

Table 6: **Trajectory Monte-Carlo compounding, numeric values.** Mean per-trajectory standard deviation of $\hat{R}_{\text{MC}}(\tau)$ over 800 Monte-Carlo realizations at $n_{\text{mc}} = 8$, fixed peaked policy π_θ (peakedness 1.0), $N_{\text{traj}} = 128$ trajectories per horizon, no training. KL is identically zero as predicted by Theorem 2, while non-KL standard deviations grow with H .

Divergence Ω	$H = 2$	$H = 4$	$H = 6$	$H = 8$
KL	0.000	0.000	0.000	0.000
α -divergence	0.32	0.51	0.78	0.95
Reverse-KL	0.84	1.38	2.29	2.81
Jensen–Shannon	0.09	0.15	0.23	0.28
squared Hellinger	0.20	0.32	0.50	0.61
χ^2	0.42	0.72	1.07	1.30
squared Euclidean	0.02	0.04	0.06	0.08

Table 7: **Recovery-quality standard deviation at fixed $N_{\text{steps}} = 500$.** Standard deviation of the per-seed value gap $\Delta(\text{root}) = \hat{V}_\theta(\text{root}) - V_\Omega^*(\text{root})$ across $N_{\text{seeds}} = 20$ training runs at horizon $H = 3$. KL is essentially flat across the sweep, while Reverse-KL and squared Hellinger drop by a factor of three from $n_{\text{mc}} = 1$ to 32.

Divergence Ω	$n_{\text{mc}} = 1$	$n_{\text{mc}} = 4$	$n_{\text{mc}} = 8$	$n_{\text{mc}} = 32$
KL	0.027	0.022	0.023	0.021
Reverse-KL	0.105	0.073	0.066	0.037
χ^2	0.036	0.022	0.018	0.026
squared Hellinger	0.161	0.106	0.102	0.049

shortfall from too few steps overtakes the variance benefit. KL avoids the tradeoff entirely because no Monte-Carlo step is needed.

B.11 DEPTH-SWEEP DIAGNOSTIC

The depth panel in Figure 13 (left) shows that all divergences, including KL, incur growing recovery standard deviation as horizon increases. We attribute this to a budget-per-state confound. Holding $N_{\text{pairs}} = 1000$ fixed while $|\mathcal{S}|$ grows as W^{H+1} amounts to a shrinking effective sample-per-state, so longer-horizon trees are under-trained at the same compute. KL has zero Monte-Carlo noise by Theorem 2 but still inherits this generic optimization difficulty. The two effects are separated by the trajectory Monte-Carlo compounding experiment in Appendix B.9, which uses no training and a fixed π_θ . In that experiment KL standard deviation is exactly zero at all H . The vocabulary-scaling experiment in Appendix B.16 also separates the two effects in the orthogonal direction by scaling the preference budget with $|\mathcal{S}|$. A budget-controlled depth sweep is left to future work.

B.12 FULL $N_{\text{pairs}} \times n_{\text{mc}}$ HEATMAP OF Δ_r

The headline curve in Figure 5 is a cross-section of a 6×9 joint sweep over $N_{\text{pairs}} \in \{500, 1\,000, 2\,000, 4\,000, 8\,000, 16\,000\}$ and $n_{\text{mc}} \in \{1, 2, 4, 8, 16, 32, 64, 128, 256\}$, for each of the seven Ω at both regimes. Figure 14 shows the full Δ_r heatmap with a single shared color scale. KL panels are nearly uniform (only the bottom row, $N_{\text{pairs}} = 500$, has visibly higher values), reflecting the theorem’s n_{mc} -invariance and a mild data-noise floor at small N_{pairs} . Reverse-KL, α -divergence, Jensen–Shannon, and squared Hellinger panels all show a diagonal gradient from purple at small $(N_{\text{pairs}}, n_{\text{mc}})$ to yellow at large; both axes reduce the gap. Pearson χ^2 and squared Euclidean panels are roughly uniform, consistent with the failure-to-converge observations in the main text. Five seeds per cell. Source: `experiments/13_npairs_nmc_joint.py`, `replot` via `experiments/13_replot_shared_scale.py`.

B.13 ECE HEATMAP WITH NONLINEAR COLOR SCALE

The same joint sweep evaluated by soft-target ECE. Because ECE values span two orders of magnitude across Ω (KL and χ^2 near 10^{-2} , JS and Euclidean on-policy near 0.25), a linear color scale hides

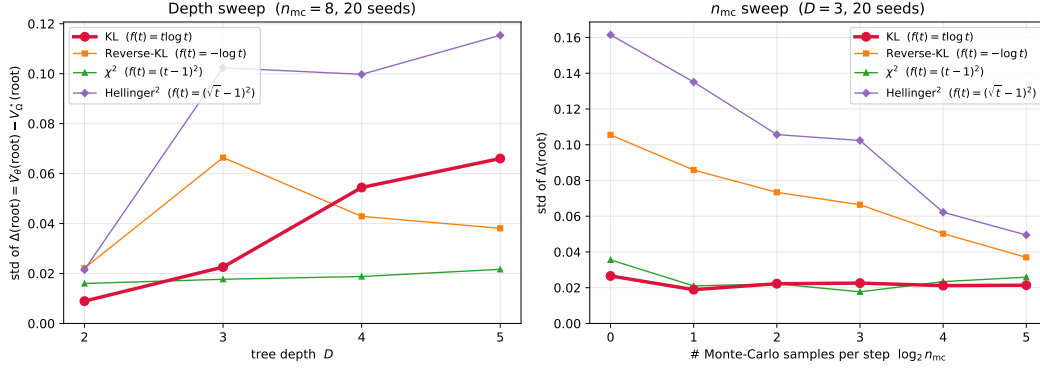


Figure 13: **Fixed- N_{steps} sweeps.** Right panel reports the n_{mc} sweep summarized in Table 7. KL (red) is essentially flat. Reverse-KL and squared Hellinger fall as n_{mc} grows. Left panel reports a depth sweep at fixed $n_{mc} = 8$ and is discussed in Appendix B.11.

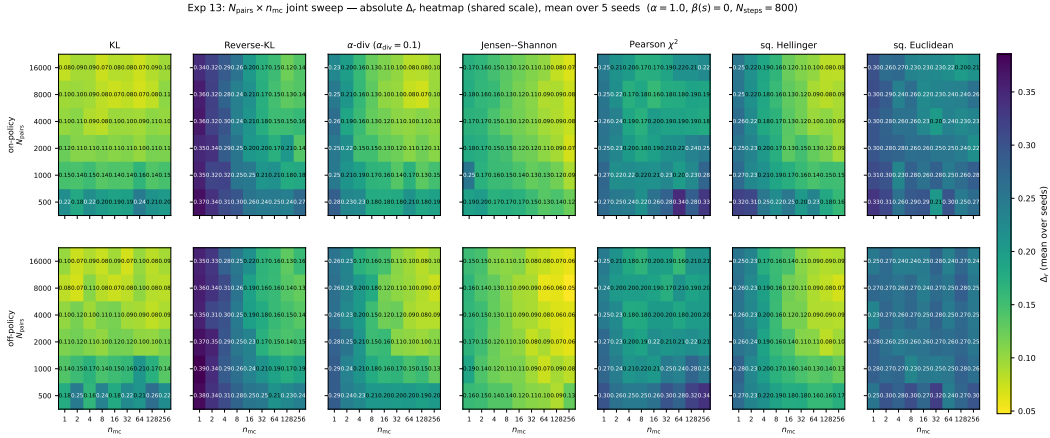


Figure 14: Equivalent-reward gap Δ_r as a function of both N_{pairs} (y) and n_{mc} (x), for 7 regularizers (cols) $\times$ on/off-policy (rows). Mean over 5 seeds, shared linear color scale across all panels.

the low-ECE variation. Figure 15 uses a PowerNorm($\gamma = 0.5$) remap (a square-root compression of the high-ECE tail) that expands the low end while keeping the high values visible. The on-policy plateaus for JS and squared Euclidean stand out as solid dark bands invariant to both axes, mirroring the on-policy ECE plateau in the main-text curve panel.

B.14 WHY PEARSON χ^2 AND SQUARED EUCLIDEAN FALL OUTSIDE THE FOUR- Ω PATTERN

The four Φ -bounded f -divergences in Section 4.3 converge toward KL’s Δ_r floor as n_{mc} grows. Pearson χ^2 and squared Euclidean do not, and for reasons that have nothing to do with inner-term variance.

Squared Euclidean is a Bregman regularizer rather than an f -divergence, so its optimal policy π_{Eucl}^* is a simplex projection of Q^* rather than a softmax. Trajectory-DPO is built around the softmax-shaped implicit reward and is misaligned with this geometry, so its Δ_r stays bounded away from the data and optimization floor at every n_{mc} .

Pearson χ^2 has $\Phi_{\chi^2}(t) = 2(t-1) - (t-1)^2/t$, which carries a $1/t$ pole as $t \rightarrow 0$. Its Monte-Carlo integrand is therefore heavy-tailed, and a non-vanishing residual survives even at $n_{mc} = 256$. A second, unrelated effect makes its gap deceptively small: at $\alpha = 1$ with a uniform π_{ref} , the quadratic penalty pulls $\pi_{\chi^2}^*$ close to π_{ref} , so the canonical reward $r_{\chi^2, \text{eq}}^*$ has little dynamic range and $\Delta_r^{\chi^2}$ can look small for trivial reasons.

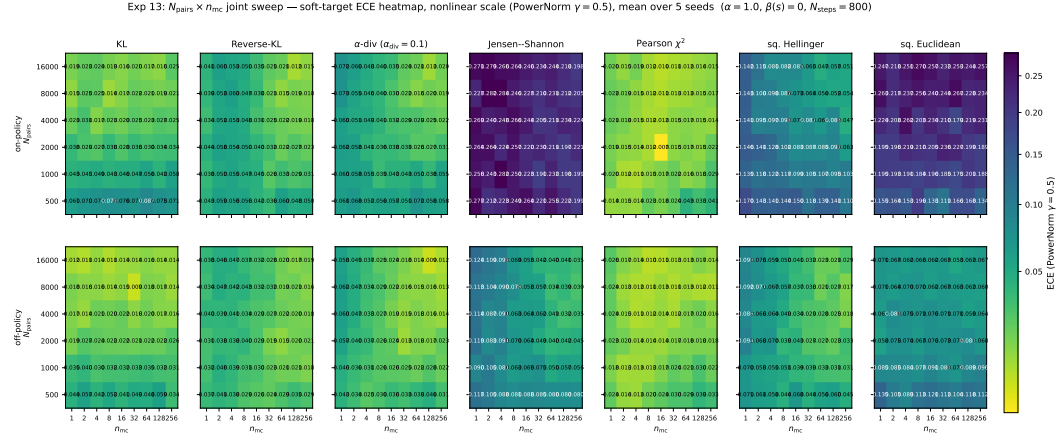


Figure 15: Soft-target ECE heatmap on the same grid as Figure 14, with a PowerNorm($\gamma = 0.5$) color remap so that low-ECE differentiation (KL, Reverse-KL, α -div, χ^2) and high-ECE plateaus (on-policy JS, squared Hellinger, Euclidean) are both legible. Mean over 5 seeds, shared remapped color scale.

B.15 COMPUTE-MATCHED RECOVERY (COMPLEMENT TO SECTION 4.3)

This appendix reports the compute-matched recovery experiment in the trained-policy quality metrics of earlier drafts. The fairness caveat raised in the lead-in to Section 4 applies: normalized regret and recovery gap each measure how well a policy trained under Ω recovers π_Ω^* , and that target itself depends on Ω . The result is best read alongside the equivalent-reward gap of Section 4.3.

Counting per-step training cost as $1 + n_{\text{mc}}$ forward passes per non-terminal transition, we hold the total per-regularizer compute budget $C = N_{\text{steps}} \times (1 + n_{\text{mc}})$ constant across Ω . KL uses C gradient steps with no inner-term sampling. Each non-KL regularizer is run separately at $n_{\text{mc}} \in \{1, 4, 16\}$, and at each compute level the column reports the best-achieved value. Compute levels $C \in \{100, 500, 1000, 5000, 10000, 50000\}$ span two and a half orders of magnitude. Five seeds per configuration.

Table 8: Compute-matched normalized regret at the root. Mean across 5 seeds at horizon $H = 3$. For each non-KL regularizer the cell reports the best-achieved value across the $n_{\text{mc}} \in \{1, 4, 16\}$ sweep at compute budget C . KL runs with no Monte-Carlo step and uses $N_{\text{steps}} = C$ gradient steps.

Divergence Ω	$C=100$	$C=500$	$C=1000$	$C=5000$	$C=10\,000$	$C=50\,000$
KL	0.11	0.12	0.11	0.10	0.11	0.10
Reverse-KL	0.41	0.29	0.24	0.24	0.25	0.23
χ^2	0.27	0.30	0.28	0.29	0.28	0.28
squared Hellinger	0.43	0.29	0.23	0.16	0.18	0.14

KL is flat at a normalized regret of roughly 0.11 across the entire compute range, while every non-KL regularizer needs orders of magnitude more compute to merely approach its own plateau and never reaches the KL line. Squared Hellinger improves the most with compute, halving its regret from 0.43 at $C = 100$ to 0.14 at $C = 50\,000$, but at the largest budget it still sits 40% above KL. Chi-squared is the slowest to improve and plateaus at roughly 0.28. The recovery-gap panel in Figure 16 confirms the same ordering in absolute leaf-reward units: KL has a recovery gap of ≈ 0.05 at all compute levels while non-KL recovery gaps range from 0.10 to 0.86.

B.16 VOCABULARY SCALING (COMPLEMENT TO SECTION 4.3)

This appendix reports the vocabulary-scaling recovery experiment in the same trained-policy quality metrics. The fairness caveat from Section 4 applies. The toy tree of Appendix B.1 has $W = 3$ actions per state; language-model post-training operates at vocabulary sizes of tens of thousands.

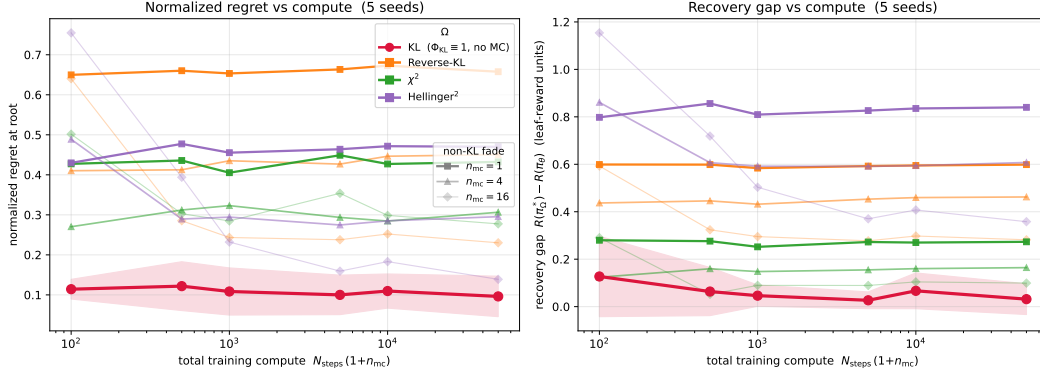


Figure 16: Trained-policy recovery as a function of total training compute. Left: normalized regret at the root. Right: recovery gap in leaf-reward units. For each non-KL divergence the three lines correspond to $n_{\text{mc}} \in \{1, 4, 16\}$, drawn at progressively lighter shades. KL (crimson) is a single line because it pays cost 1 per step. Shaded bands are ± 1 seed-level standard deviation across 5 seeds.

The per-state inner-term variance grows with $|\mathcal{A}|$ because the Monte-Carlo support widens and Φ_{Ω} picks up more low-probability arguments. We vary W at fixed horizon $H = 2$ on a log grid $W \in \{3, 10, 30, 100, 300, 1000, 3000, 5000\}$. Preference budget scales linearly with W (capped at 50 000 to bound memory), $n_{\text{mc}} = 8$, $N_{\text{steps}} = 800$, three seeds.

Table 9: Vocabulary-scaling recovery gap at the root, in leaf-reward units. Mean across 3 seeds at horizon $H = 2$, $n_{\text{mc}} = 8$.

Ω	$W=3$	$W=10$	$W=30$	$W=100$	$W=300$	$W=1\,000$	$W=3\,000$	$W=5\,000$
KL	-0.02	0.00	0.01	0.46	0.79	0.97	1.00	1.00
Reverse-KL	0.17	0.54	0.85	1.43	2.01	2.52	3.02	3.02
χ^2	0.04	0.09	0.06	0.17	0.36	0.47	0.51	0.49
sq. Hellinger	0.15	0.79	1.23	2.21	3.22	3.89	4.45	4.38

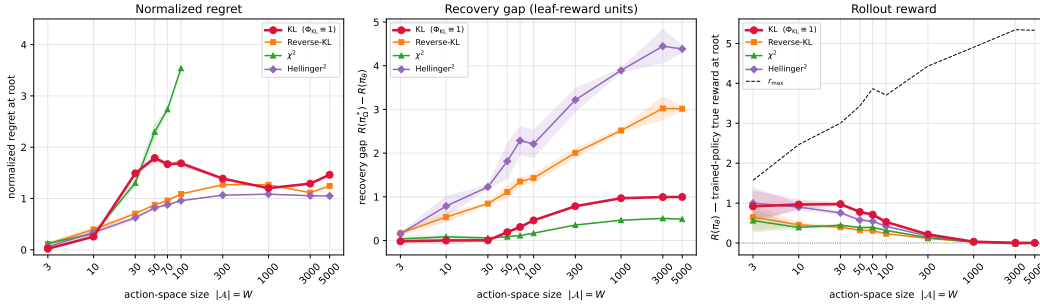


Figure 17: Vocabulary scaling of trained-policy recovery, three views. Left: normalized regret. Middle: recovery gap in leaf-reward units. Right: trained-policy rollout reward $R(\pi_{\theta})$, evaluated by running π_{θ} in the tree and reading the leaf reward (i.e., $V^{\pi_{\theta}}$ at $\alpha = 0$). The black dashed line on the right panel is r_{max} , the best leaf reward; the dotted line at zero is $R(\pi_{\text{ref}})$. Horizon $H = 2$, $n_{\text{mc}} = 8$, $N_{\text{steps}} = 800$, three seeds. Within each (seed, W) cell, all four Ω share the same leaf-reward draw and the same Bradley–Terry preference set.

KL keeps a near-zero recovery gap up to $W = 30$ and plateaus near 1.0 leaf-reward unit at $W \geq 1000$; squared Hellinger and Reverse-KL grow monotonically and reach 4.4 and 3.0 at $W = 5000$, three to four times the KL value. Chi-squared has a small recovery gap throughout, but for a reason orthogonal to the inner-term variance mechanism: at $\alpha = 1$ the chi-squared penalty pulls $\pi_{\chi^2}^*$ very close to π_{ref} , so the trained policy has a target that is itself near the reference and the gap stays small even when

the policy barely moves from π_{ref} . Reverse-KL and squared Hellinger admit more concentrated π_{Ω}^* and so expose the predicted variance-driven recovery degradation directly. The rollout panel (right) makes the underlying training dynamics explicit: for $W \leq 30$ KL recovers a policy with $R(\pi_{\theta}) \approx 0.98$, within a few percent of its own ceiling $R(\pi_{\text{KL}}^*) \approx 1.0$; from $W = 50$ onward $R(\pi_{\theta})$ for every Ω slides toward 0 because at this scale the tabular policy class has more parameters than the fixed compute budget can fit. The overall pattern is consistent with Theorem 2, with the caveat that at very large W the toy tabular setup is in the under-parameterized regime for every Ω ; a parametric LM-scale follow-up that avoids this is a natural next step.

B.17 PER-LEAF ANATOMY: OFF-POLICY AND NON-UNIFORM π_{ref} COUNTERPARTS

The main-text per-leaf anatomy (Figure 6) shows the on-policy regime under uniform π_{ref} . This appendix collects the three companion panels: off-policy with uniform π_{ref} (Figure 18), and both regimes with the non-uniform π_{ref} from Appendix B.7 (Figures 19 and 20). All four use the same calibrated α_{Ω}^* as the main text, and the panel conventions are identical: coloured solid = π_{θ} , black dashed = that Ω ’s own π_{Ω}^* . The off-policy regime narrows the per-leaf gap for most non-KL Ω because uniform-reference rollouts cover off-modal leaves more evenly, while KL remains essentially unchanged. The non-uniform π_{ref} shifts the shape of π_{Ω}^* but preserves the cross- Ω pattern: KL closes the per-leaf gap; the six non-KL Ω leave the same characteristic residuals.

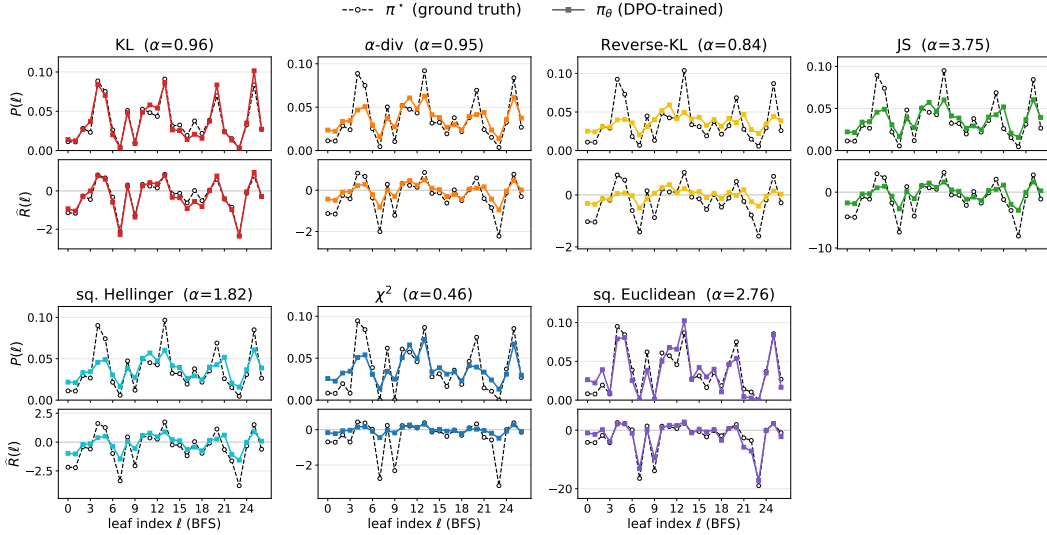


Figure 18: **Off-policy counterpart of Figure 6, behavior = π_{ref} (uniform).**

B.18 OFF-POLICY COUNTERPART OF FIGURE 7 AND THE α -DIVERGENCE RE-NORMALIZATION

The main-text α -divergence sweep (Section 4.5, Figure 7) uses the family $f_{\rho}(t) = (t^{1-\rho} - (1-\rho)t - \rho)/(\rho(\rho-1))$ at $\rho \in \{0.01, 0.25, 0.50, 0.75, 0.99\}$, which interpolates between KL ($\rho \rightarrow 0$) and Reverse-KL ($\rho \rightarrow 1$) through a Hellinger-like midpoint at $\rho = 0.5$. To make the limits coincide with the native KL and Reverse-KL implementations used elsewhere in the paper, we re-normalize f_{ρ} by the linear-in- t correction $(1-2\rho)(t-1)$, which is divergence-preserving (linear-in- t shifts integrate to a constant over the simplex) but pins the endpoint per- (s, a) values of $\nabla\Omega$ to the project conventions. Without this re-normalization, the inner-term integrand at $\rho \rightarrow 1$ differs from the native Reverse-KL one by a state-dependent term whose Monte-Carlo expectation is the constant 1 (a no-op in the divergence functional) but whose $n_{\text{mc}} = 1$ variance is non-zero, so the trained policy at $\rho \rightarrow 1$ would drift away from the policy obtained with the native implementation even though the underlying optimal π^* matches. After re-normalization the $\rho = 0.01$ trained policy matches the native KL trained policy to within numerical noise (maximum per- (s, a) deviation < 0.01) and analogously for $\rho = 0.99$ versus native Reverse-KL.

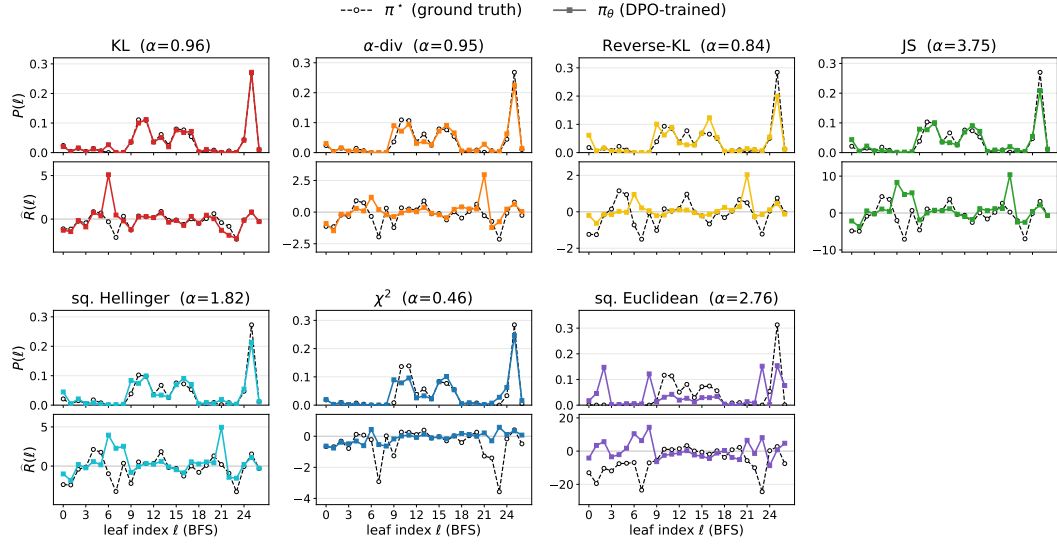
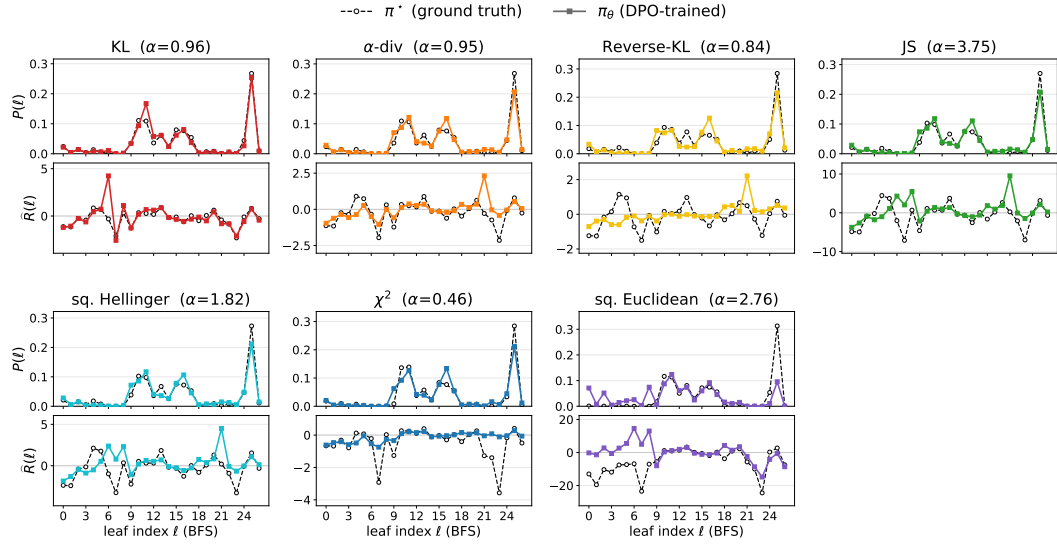
Figure 19: Per-leaf anatomy, on-policy, non-uniform π_{ref} (peakedness 1.0, seed 999).Figure 20: Off-policy counterpart of Figure 19, non-uniform π_{ref} .

Figure 21 shows the off-policy counterpart of the main-text sweep. The fan opens less dramatically than on-policy because the uniform rollouts give all five trained policies more uniform coverage of off-modal leaves, partially neutralizing the $n_{\text{mc}} = 1$ inner-term variance differences across ρ .

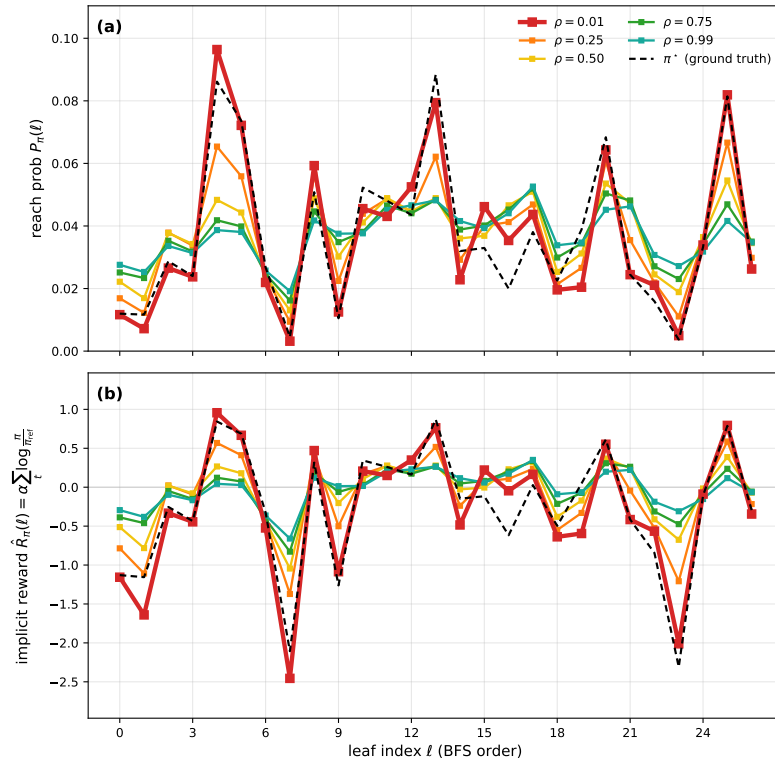


Figure 21: Off-policy counterpart of Figure 7, behavior = π_{ref} (uniform). Same axes and conventions as the main-text figure.